Supplementary Information

Translate or encode? Two design cultures for coupling language and omics data in generative foundation models: a scoping review.

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Supplementary items are numbered by type in the order in which the article first cites them. Numbers in square brackets refer to the article's reference list. Data files are listed at the end with their column dictionary.

Supplementary Notes

Supplementary Note S1. Search strategies as executed

Every query string as executed, by round and source, with the execution date and the number of records retained from each source. The strings are reproduced verbatim from the search configurations.

Round 1, baseline (protocol version 3.1): 2018-01-01 to 2026-02-28

PubMed (executed 2026-02-15; 631 records retained)

```
(("single cell"[Title/Abstract] OR "single-cell"[Title/Abstract] OR scRNA-seq[Title/Abstract] OR
↪  "RNA-seq"[Title/Abstract] OR "RNA seq"[Title/Abstract] OR "gene expression"[Title/Abstract] OR
↪  scATAC-seq[Title/Abstract] OR "ATAC-seq"[Title/Abstract] OR "chromatin accessibility"[Title/Abstract] OR
↪  CITE-seq[Title/Abstract] OR "spatial transcriptomics"[Title/Abstract] OR "multi-omics"[Title/Abstract] OR "multi
↪  omics"[Title/Abstract] OR multiomics[Title/Abstract] OR transcriptomics[Title/Abstract] OR
↪  genomics[Title/Abstract] OR proteomics[Title/Abstract] OR epigenomics[Title/Abstract] OR "cell
↪  type"[Title/Abstract] OR "gene regulatory"[Title/Abstract]) AND ("language model"[Title/Abstract] OR
↪  LLM[Title/Abstract] OR "large language model"[Title/Abstract] OR "natural language"[Title/Abstract] OR
↪  GPT[Title/Abstract] OR CLIP[Title/Abstract] OR "cross-modal"[Title/Abstract] OR multimodal[Title/Abstract] OR
↪  "multi-modal"[Title/Abstract] OR tokenization[Title/Abstract] OR "gene token"[Title/Abstract] OR
↪  prompt[Title/Abstract] OR "text generation"[Title/Abstract] OR "cell-to-text"[Title/Abstract] OR
↪  "text-to-cell"[Title/Abstract] OR NLP[Title/Abstract]) AND ("foundation model"[Title/Abstract] OR
↪  pretrained[Title/Abstract] OR "pre-trained"[Title/Abstract] OR "self-supervised"[Title/Abstract] OR
↪  transformer[Title/Abstract] OR generative[Title/Abstract] OR decoder[Title/Abstract] OR
↪  autoregressive[Title/Abstract] OR "transfer learning"[Title/Abstract] OR "zero-shot"[Title/Abstract] OR
↪  "few-shot"[Title/Abstract] OR "masked modeling"[Title/Abstract] OR "attention mechanism"[Title/Abstract] OR "deep
↪  learning"[Title/Abstract])) AND ("2018/01/01"[Date - Publication] : "2026/02/28"[Date - Publication]) AND "free
↪  full text"[sb] AND English[Language]
```

Scopus (executed 2026-02-15; 1,021 records retained)

```
TITLE-ABS-KEY(("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR
↪  scATAC-seq OR "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR
↪  "multi omics" OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene
↪  regulatory") AND ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR
↪  "cross-modal" OR multimodal OR "multi-modal" OR tokeniz* OR "gene token" OR prompt OR "text generation" OR
↪  "cell-to-text" OR "text-to-cell" OR NLP) AND ("foundation model" OR pretrain* OR "pre-train*" OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "masked modeling" OR "attention mechanism" OR "deep learning")) AND PUBYEAR > 2017 AND OPENACCESS(1) AND
↪  LANGUAGE(English)
```

Semantic Scholar (executed 2026-02-15; 2,192 records retained)

```
("single-cell" | "single cell" | "scRNA-seq" | "RNA-seq" | "RNA seq" | "gene expression" | transcriptomics | genomics
↪  | proteomics | epigenomics | "spatial transcriptomics" | "multi-omics" | "multi omics") + ("language model" | LLM
↪  | "large language model" | "natural language" | GPT | CLIP | multimodal | "multi-modal" | "cross-modal" | tokeniz*
↪  | prompt | NLP | "text generation") + ("foundation model" | pretrain* | "self-supervised" | transformer |
↪  generative | decoder | autoregressive | "transfer learning" | "zero-shot" | "deep learning")
```

arXiv (executed 2026-02-15; 187 records retained)

```
[primary] (ti:"single cell" OR ti:scRNA OR ti:"gene expression" OR ti:transcriptomics OR ti:genomics OR
↪  abs:"single-cell" OR abs:"RNA-seq" OR abs:"RNA seq" OR abs:"spatial transcriptomics" OR abs:"multi-omics" OR
↪  abs:"multi omics") AND (ti:"language model" OR ti:GPT OR ti:LLM OR ti:CLIP OR ti:multimodal OR abs:"language
↪  model" OR abs:"large language model" OR abs:"natural language" OR abs:"cross-modal" OR abs:"multi-modal" OR
↪  abs:tokeniz OR abs:prompt) AND (abs:"foundation model" OR abs:pretrain OR abs:"self-supervised" OR abs:transformer
↪  OR abs:generative OR abs:decoder OR abs:autoregressive OR abs:"transfer learning")
[cell_language] abs:"foundation model" AND abs:"single cell" AND abs:"language model"
[model_names] (ti:scGPT OR ti:tGPT OR ti:LangCell OR ti:ChatCell OR ti:GenePT OR ti:GeneGPT OR ti:CellPLM OR
↪  ti:Nicheformer OR ti:EpiAgent)
[bio_llm] abs:"gene expression" AND abs:LLM AND (abs:generative OR abs:decoder OR abs:autoregressive)
```

bioRxiv and medRxiv (Europe PMC preprints) (executed 2026-02-15; 669 records retained)

```
("single-cell" OR "single cell" OR "scRNA-seq" OR "RNA-seq" OR "RNA seq" OR "gene expression" OR "scATAC-seq" OR
↪  "ATAC-seq" OR "chromatin accessibility" OR "CITE-seq" OR "spatial transcriptomics" OR "multi-omics" OR "multi
↪  omics" OR transcriptomics OR genomics OR proteomics OR epigenomics) AND ("language model" OR LLM OR "large
↪  language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR multimodal OR "multi-modal" OR tokeniz*
↪  OR "gene token" OR prompt OR "text generation" OR NLP) AND ("foundation model" OR pretrain* OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "deep learning") AND (SRC:PPR) AND FIRST_PDATE:[2018-01-01 TO 2026-02-28]
```

Springer Nature (executed 2026-02-15; 272 records retained)

```
("single-cell" OR "single cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪  "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪  OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪  ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪  multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR NLP) AND
↪  ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer OR generative OR decoder OR
↪  autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked modeling" OR "attention mechanism"
↪  OR "deep learning")
```

Google Scholar (executed 2026-02-15; 562 records retained)

```
"foundation model" "gene expression" "language model" single-cell
GPT "single cell" "gene expression" generative
"natural language" "single-cell" "cell type" foundation model
multimodal LLM genomics transcriptomics "foundation model"
scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR GeneGPT OR CellPLM OR Nicheformer OR EpiAgent
"multi omics" "language model" "foundation model" transformer
"RNA seq" LLM "foundation model" generative single-cell
```

Round 2, April top-up: 2026-03-01 to 2026-04-14

PubMed (executed 2026-04-14; 46 records retained)

```
(("single cell"[Title/Abstract] OR "single-cell"[Title/Abstract] OR scRNA-seq[Title/Abstract] OR
↪  "RNA-seq"[Title/Abstract] OR "RNA seq"[Title/Abstract] OR "gene expression"[Title/Abstract] OR
↪  scATAC-seq[Title/Abstract] OR "ATAC-seq"[Title/Abstract] OR "chromatin accessibility"[Title/Abstract] OR
↪  CITE-seq[Title/Abstract] OR "spatial transcriptomics"[Title/Abstract] OR "multi-omics"[Title/Abstract] OR "multi
↪  omics"[Title/Abstract] OR multiomics[Title/Abstract] OR transcriptomics[Title/Abstract] OR
↪  genomics[Title/Abstract] OR proteomics[Title/Abstract] OR epigenomics[Title/Abstract] OR "cell
↪  type"[Title/Abstract] OR "gene regulatory"[Title/Abstract]) AND ("language model"[Title/Abstract] OR
↪  LLM[Title/Abstract] OR "large language model"[Title/Abstract] OR "natural language"[Title/Abstract] OR
↪  GPT[Title/Abstract] OR CLIP[Title/Abstract] OR "cross-modal"[Title/Abstract] OR multimodal[Title/Abstract] OR
↪  "multi-modal"[Title/Abstract] OR tokenization[Title/Abstract] OR "gene token"[Title/Abstract] OR
↪  prompt[Title/Abstract] OR "text generation"[Title/Abstract] OR "cell-to-text"[Title/Abstract] OR
↪  "text-to-cell"[Title/Abstract] OR NLP[Title/Abstract]) AND ("foundation model"[Title/Abstract] OR
↪  pretrained[Title/Abstract] OR "pre-trained"[Title/Abstract] OR "self-supervised"[Title/Abstract] OR
↪  transformer[Title/Abstract] OR generative[Title/Abstract] OR decoder[Title/Abstract] OR
↪  autoregressive[Title/Abstract] OR "transfer learning"[Title/Abstract] OR "zero-shot"[Title/Abstract] OR
↪  "few-shot"[Title/Abstract] OR "masked modeling"[Title/Abstract] OR "attention mechanism"[Title/Abstract] OR "deep
↪  learning"[Title/Abstract])) AND ("2026/03/01"[Date - Publication] : "2026/04/14"[Date - Publication]) AND "free
↪  full text"[sb] AND English[Language]
```

Scopus (executed 2026-04-14; 32 records retained)

```
TITLE-ABS-KEY(("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR
↪  scATAC-seq OR "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR
↪  "multi omics" OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene
↪  regulatory") AND ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR
↪  "cross-modal" OR multimodal OR "multi-modal" OR tokeniz* OR "gene token" OR prompt OR "text generation" OR
↪  "cell-to-text" OR "text-to-cell" OR NLP) AND ("foundation model" OR pretrain* OR "pre-train*" OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "masked modeling" OR "attention mechanism" OR "deep learning")) AND PUBYEAR > 2025 AND OPENACCESS(1) AND
↪  LANGUAGE(English)
```

Semantic Scholar (executed 2026-04-14; 201 records retained)

```
("single-cell" | "single cell" | "scRNA-seq" | "RNA-seq" | "RNA seq" | "gene expression" | transcriptomics | genomics
↪  | proteomics | epigenomics | "spatial transcriptomics" | "multi-omics" | "multi omics") + ("language model" | LLM
↪  | "large language model" | "natural language" | GPT | CLIP | multimodal | "multi-modal" | "cross-modal" | tokeniz*
↪  | prompt | NLP | "text generation") + ("foundation model" | pretrain* | "self-supervised" | transformer |
↪  generative | decoder | autoregressive | "transfer learning" | "zero-shot" | "deep learning")
```

arXiv (executed 2026-04-14; 14 records retained)

```
[primary] (ti:"single cell" OR ti:scRNA OR ti:"gene expression" OR ti:transcriptomics OR ti:genomics OR
↪  abs:"single-cell" OR abs:"RNA-seq" OR abs:"RNA seq" OR abs:"spatial transcriptomics" OR abs:"multi-omics" OR
↪  abs:"multi omics") AND (ti:"language model" OR ti:GPT OR ti:LLM OR ti:CLIP OR ti:multimodal OR abs:"language
↪  model" OR abs:"large language model" OR abs:"natural language" OR abs:"cross-modal" OR abs:"multi-modal" OR
↪  abs:tokeniz OR abs:prompt) AND (abs:"foundation model" OR abs:pretrain OR abs:"self-supervised" OR abs:transformer
↪  OR abs:generative OR abs:decoder OR abs:autoregressive OR abs:"transfer learning")
[cell_language] abs:"foundation model" AND abs:"single cell" AND abs:"language model"
[model_names] (ti:scGPT OR ti:tGPT OR ti:LangCell OR ti:ChatCell OR ti:GenePT OR ti:GeneGPT OR ti:CellPLM OR
↪  ti:Nicheformer OR ti:EpiAgent)
[bio_llm] abs:"gene expression" AND abs:LLM AND (abs:generative OR abs:decoder OR abs:autoregressive)
```

bioRxiv and medRxiv (Europe PMC preprints) (executed 2026-04-14; 5 records retained)

```
("single-cell" OR "single cell" OR "scRNA-seq" OR "RNA-seq" OR "RNA seq" OR "gene expression" OR "scATAC-seq" OR
↪  "ATAC-seq" OR "chromatin accessibility" OR "CITE-seq" OR "spatial transcriptomics" OR "multi-omics" OR "multi
↪  omics" OR transcriptomics OR genomics OR proteomics OR epigenomics) AND ("language model" OR LLM OR "large
↪  language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR multimodal OR "multi-modal" OR tokeniz*
↪  OR "gene token" OR prompt OR "text generation" OR NLP) AND ("foundation model" OR pretrain* OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "deep learning") AND (SRC:PPR) AND FIRST_PDATE:[2026-03-01 TO 2026-04-14]
```

Springer Nature (executed 2026-04-14; 33 records retained)

```
("single-cell" OR "single cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪  "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪  OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪  ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪  multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR NLP) AND
↪  ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer OR generative OR decoder OR
↪  autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked modeling" OR "attention mechanism"
↪  OR "deep learning")
```

Google Scholar (executed 2026-04-14; 536 records retained)

```
"foundation model" "gene expression" "language model" single-cell
GPT "single cell" "gene expression" generative
"natural language" "single-cell" "cell type" foundation model
multimodal LLM genomics transcriptomics "foundation model"
scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR GeneGPT OR CellPLM OR Nicheformer OR EpiAgent
"multi omics" "language model" "foundation model" transformer
"RNA seq" LLM "foundation model" generative single-cell
```

Notes logged with the round: Semantic Scholar API key expired (HTTP 403), used keyless access. Google Scholar rate-limited after query 5/7.

Round 3, June top-up: 2026-04-15 to 2026-06-10

PubMed (executed 2026-06-10; 68 records retained)

```
(("single cell"[Title/Abstract] OR "single-cell"[Title/Abstract] OR scRNA-seq[Title/Abstract] OR
↪  "RNA-seq"[Title/Abstract] OR "RNA seq"[Title/Abstract] OR "gene expression"[Title/Abstract] OR
↪  scATAC-seq[Title/Abstract] OR "ATAC-seq"[Title/Abstract] OR "chromatin accessibility"[Title/Abstract] OR
↪  CITE-seq[Title/Abstract] OR "spatial transcriptomics"[Title/Abstract] OR "multi-omics"[Title/Abstract] OR "multi
↪  omics"[Title/Abstract] OR multiomics[Title/Abstract] OR transcriptomics[Title/Abstract] OR
↪  genomics[Title/Abstract] OR proteomics[Title/Abstract] OR epigenomics[Title/Abstract] OR "cell
↪  type"[Title/Abstract] OR "gene regulatory"[Title/Abstract]) AND ("language model"[Title/Abstract] OR
↪  LLM[Title/Abstract] OR "large language model"[Title/Abstract] OR "natural language"[Title/Abstract] OR
↪  GPT[Title/Abstract] OR CLIP[Title/Abstract] OR "cross-modal"[Title/Abstract] OR multimodal[Title/Abstract] OR
↪  "multi-modal"[Title/Abstract] OR tokenization[Title/Abstract] OR "gene token"[Title/Abstract] OR
↪  prompt[Title/Abstract] OR "text generation"[Title/Abstract] OR "cell-to-text"[Title/Abstract] OR
↪  "text-to-cell"[Title/Abstract] OR NLP[Title/Abstract]) AND ("foundation model"[Title/Abstract] OR
↪  pretrained[Title/Abstract] OR "pre-trained"[Title/Abstract] OR "self-supervised"[Title/Abstract] OR
↪  transformer[Title/Abstract] OR generative[Title/Abstract] OR decoder[Title/Abstract] OR
↪  autoregressive[Title/Abstract] OR "transfer learning"[Title/Abstract] OR "zero-shot"[Title/Abstract] OR
↪  "few-shot"[Title/Abstract] OR "masked modeling"[Title/Abstract] OR "attention mechanism"[Title/Abstract] OR "deep
↪  learning"[Title/Abstract])) AND ("2026/04/15"[Date - Publication] : "2026/06/10"[Date - Publication]) AND "free
↪  full text"[sb] AND English[Language]
```

Scopus (executed 2026-06-10; 39 records retained)

```
TITLE-ABS-KEY(("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR
↪ scATAC-seq OR "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR
↪ "multi omics" OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene
↪ regulatory") AND ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR
↪ "cross-modal" OR multimodal OR "multi-modal" OR tokeniz* OR "gene token" OR prompt OR "text generation" OR
↪ "cell-to-text" OR "text-to-cell" OR NLP) AND ("foundation model" OR pretrain* OR "pre-train*" OR "self-supervised"
↪ OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪ "masked modeling" OR "attention mechanism" OR "deep learning")) AND PUBYEAR > 2025 AND OPENACCESS(1) AND
↪ LANGUAGE(English)
```

Semantic Scholar (executed 2026-06-10; 215 records retained)

```
("single-cell" | "single cell" | "scRNA-seq" | "RNA-seq" | "RNA seq" | "gene expression" | transcriptomics | genomics
↪ | proteomics | epigenomics | "spatial transcriptomics" | "multi-omics" | "multi omics") + ("language model" | LLM
↪ | "large language model" | "natural language" | GPT | CLIP | multimodal | "multi-modal" | "cross-modal" | tokeniz*
↪ | prompt | NLP | "text generation") + ("foundation model" | pretrain* | "self-supervised" | transformer |
↪ generative | decoder | autoregressive | "transfer learning" | "zero-shot" | "deep learning")
```

arXiv (executed 2026-06-10; 21 records retained)

```
[primary] (ti:"single cell" OR ti:scRNA OR ti:"gene expression" OR ti:transcriptomics OR ti:genomics OR
↪ abs:"single-cell" OR abs:"RNA-seq" OR abs:"RNA seq" OR abs:"spatial transcriptomics" OR abs:"multi-omics" OR
↪ abs:"multi omics") AND (ti:"language model" OR ti:GPT OR ti:LLM OR ti:CLIP OR ti:multimodal OR abs:"language
↪ model" OR abs:"large language model" OR abs:"natural language" OR abs:"cross-modal" OR abs:"multi-modal" OR
↪ abs:tokeniz OR abs:prompt) AND (abs:"foundation model" OR abs:pretrain OR abs:"self-supervised" OR abs:transformer
↪ OR abs:generative OR abs:decoder OR abs:autoregressive OR abs:"transfer learning")
[cell_language] abs:"foundation model" AND abs:"single cell" AND abs:"language model"
[model_names] (ti:scGPT OR ti:tGPT OR ti:LangCell OR ti:ChatCell OR ti:GenePT OR ti:GeneGPT OR ti:CellPLM OR
↪ ti:Nicheformer OR ti:EpiAgent)
[bio_llm] abs:"gene expression" AND abs:LLM AND (abs:generative OR abs:decoder OR abs:autoregressive)
```

bioRxiv and medRxiv (Europe PMC preprints) (executed 2026-06-10; 33 records retained)

```
("single-cell" OR "single cell" OR "scRNA-seq" OR "RNA-seq" OR "RNA seq" OR "gene expression" OR "scATAC-seq" OR
↪ "ATAC-seq" OR "chromatin accessibility" OR "CITE-seq" OR "spatial transcriptomics" OR "multi-omics" OR "multi
↪ omics" OR transcriptomics OR genomics OR proteomics OR epigenomics) AND ("language model" OR LLM OR "large
↪ language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR multimodal OR "multi-modal" OR tokeniz*
↪ OR "gene token" OR prompt OR "text generation" OR NLP) AND ("foundation model" OR pretrain* OR "self-supervised"
↪ OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪ "deep learning") AND (SRC:PPR) AND FIRST_PDATE:[2026-04-15 TO 2026-06-10]
```

Springer Nature (executed 2026-06-10; 27 records retained)

```
("single-cell" OR "single cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪ "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪ OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪ ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪ multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR NLP) AND
↪ ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer OR generative OR decoder OR
↪ autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked modeling" OR "attention mechanism"
↪ OR "deep learning")
```

Google Scholar (executed 2026-06-10; 530 records retained)

```
"foundation model" "gene expression" "language model" single-cell
GPT "single cell" "gene expression" generative
"natural language" "single-cell" "cell type" foundation model
multimodal LLM genomics transcriptomics "foundation model"
scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR GeneGPT OR CellPLM OR Nicheformer OR EpiAgent
"multi omics" "language model" "foundation model" transformer
"RNA seq" LLM "foundation model" generative single-cell
```

Notes logged with the round: Top-up search after 2026-04-14. PubMed ESearch required POST with retries after transient NCBI server errors (HTTP 500). Google Scholar rate-limited after query 5/7; queries 6-7 skipped.

Round 4, July top-up: 2026-06-11 to 2026-07-06

PubMed (executed 2026-07-06; 25 records retained)

```
(("single cell"[Title/Abstract] OR "single-cell"[Title/Abstract] OR scRNA-seq[Title/Abstract] OR
↪  "RNA-seq"[Title/Abstract] OR "RNA seq"[Title/Abstract] OR "gene expression"[Title/Abstract] OR
↪  scATAC-seq[Title/Abstract] OR "ATAC-seq"[Title/Abstract] OR "chromatin accessibility"[Title/Abstract] OR
↪  CITE-seq[Title/Abstract] OR "spatial transcriptomics"[Title/Abstract] OR "multi-omics"[Title/Abstract] OR "multi
↪  omics"[Title/Abstract] OR multiomics[Title/Abstract] OR transcriptomics[Title/Abstract] OR
↪  genomics[Title/Abstract] OR proteomics[Title/Abstract] OR epigenomics[Title/Abstract] OR "cell
↪  type"[Title/Abstract] OR "gene regulatory"[Title/Abstract]) AND ("language model"[Title/Abstract] OR
↪  LLM[Title/Abstract] OR "large language model"[Title/Abstract] OR "natural language"[Title/Abstract] OR
↪  GPT[Title/Abstract] OR CLIP[Title/Abstract] OR "cross-modal"[Title/Abstract] OR multimodal[Title/Abstract] OR
↪  "multi-modal"[Title/Abstract] OR tokenization[Title/Abstract] OR "gene token"[Title/Abstract] OR
↪  prompt[Title/Abstract] OR "text generation"[Title/Abstract] OR "cell-to-text"[Title/Abstract] OR
↪  "text-to-cell"[Title/Abstract] OR NLP[Title/Abstract]) AND ("foundation model"[Title/Abstract] OR
↪  pretrained[Title/Abstract] OR "pre-trained"[Title/Abstract] OR "self-supervised"[Title/Abstract] OR
↪  transformer[Title/Abstract] OR generative[Title/Abstract] OR decoder[Title/Abstract] OR
↪  autoregressive[Title/Abstract] OR "transfer learning"[Title/Abstract] OR "zero-shot"[Title/Abstract] OR
↪  "few-shot"[Title/Abstract] OR "masked modeling"[Title/Abstract] OR "attention mechanism"[Title/Abstract] OR "deep
↪  learning"[Title/Abstract])) AND ("2026/06/11"[Date - Publication] : "2026/07/06"[Date - Publication]) AND "free
↪  full text"[sb] AND English[Language]
```

Scopus (executed 2026-07-06; 12 records retained)

```
TITLE-ABS-KEY(("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR
↪  scATAC-seq OR "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR
↪  "multi omics" OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene
↪  regulatory") AND ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR
↪  "cross-modal" OR multimodal OR "multi-modal" OR tokeniz* OR "gene token" OR prompt OR "text generation" OR
↪  "cell-to-text" OR "text-to-cell" OR NLP) AND ("foundation model" OR pretrain* OR "pre-train*" OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "masked modeling" OR "attention mechanism" OR "deep learning")) AND PUBYEAR > 2025 AND OPENACCESS(1) AND
↪  LANGUAGE(English)
```

Semantic Scholar (executed 2026-07-06; 118 records retained)

```
("single-cell" | "single cell" | "scRNA-seq" | "RNA-seq" | "RNA seq" | "gene expression" | transcriptomics | genomics
↪  | proteomics | epigenomics | "spatial transcriptomics" | "multi-omics" | "multi omics") + ("language model" | LLM
↪  | "large language model" | "natural language" | GPT | CLIP | multimodal | "multi-modal" | "cross-modal" | tokeniz*
↪  | prompt | NLP | "text generation") + ("foundation model" | pretrain* | "self-supervised" | transformer |
↪  generative | decoder | autoregressive | "transfer learning" | "zero-shot" | "deep learning")
```

arXiv (executed 2026-07-06; 6 records retained)

```
[primary] (ti:"single cell" OR ti:scRNA OR ti:"gene expression" OR ti:transcriptomics OR ti:genomics OR
↪  abs:"single-cell" OR abs:"RNA-seq" OR abs:"RNA seq" OR abs:"spatial transcriptomics" OR abs:"multi-omics" OR
↪  abs:"multi omics") AND (ti:"language model" OR ti:GPT OR ti:LLM OR ti:CLIP OR ti:multimodal OR abs:"language
↪  model" OR abs:"large language model" OR abs:"natural language" OR abs:"cross-modal" OR abs:"multi-modal" OR
↪  abs:tokeniz OR abs:prompt) AND (abs:"foundation model" OR abs:pretrain OR abs:"self-supervised" OR abs:transformer
↪  OR abs:generative OR abs:decoder OR abs:autoregressive OR abs:"transfer learning")
[cell_language] abs:"foundation model" AND abs:"single cell" AND abs:"language model"
[model_names] (ti:scGPT OR ti:tGPT OR ti:LangCell OR ti:ChatCell OR ti:GenePT OR ti:GeneGPT OR ti:CellPLM OR
↪  ti:Nicheformer OR ti:EpiAgent)
[bio_llm] abs:"gene expression" AND abs:LLM AND (abs:generative OR abs:decoder OR abs:autoregressive)
```

bioRxiv and medRxiv (Europe PMC preprints) (executed 2026-07-06; 16 records retained)

```
("single-cell" OR "single cell" OR "scRNA-seq" OR "RNA-seq" OR "RNA seq" OR "gene expression" OR "scATAC-seq" OR
↪  "ATAC-seq" OR "chromatin accessibility" OR "CITE-seq" OR "spatial transcriptomics" OR "multi-omics" OR "multi
↪  omics" OR transcriptomics OR genomics OR proteomics OR epigenomics) AND ("language model" OR LLM OR "large
↪  language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR multimodal OR "multi-modal" OR tokeniz*
↪  OR "gene token" OR prompt OR "text generation" OR NLP) AND ("foundation model" OR pretrain* OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "deep learning") AND (SRC:PPR) AND FIRST_PDATE:[2026-06-11 TO 2026-07-06]
```

Springer Nature (executed 2026-07-06; 20 records retained)

```
("single-cell" OR "single cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪  "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪  OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪  ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪  multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR NLP) AND
↪  ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer OR generative OR decoder OR
↪  autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked modeling" OR "attention mechanism"
↪  OR "deep learning")
```

Google Scholar (executed 2026-07-06; 0 records retained)

```
"foundation model" "gene expression" "language model" single-cell
GPT "single cell" "gene expression" generative
"natural language" "single-cell" "cell type" foundation model
multimodal LLM genomics transcriptomics "foundation model"
scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR GeneGPT OR CellPLM OR Nicheformer OR EpiAgent
"multi omics" "language model" "foundation model" transformer
"RNA seq" LLM "foundation model" generative single-cell
```

Round 5, August living update (protocol version 3.3): 2026-07-07 to 2026-08-09

PubMed (executed 2026-08-10; 27 records retained)

```
(("single cell"[Title/Abstract] OR "single-cell"[Title/Abstract] OR scRNA-seq[Title/Abstract] OR
↪  "RNA-seq"[Title/Abstract] OR "RNA seq"[Title/Abstract] OR "gene expression"[Title/Abstract] OR
↪  scATAC-seq[Title/Abstract] OR "ATAC-seq"[Title/Abstract] OR "chromatin accessibility"[Title/Abstract] OR
↪  CITE-seq[Title/Abstract] OR "spatial transcriptomics"[Title/Abstract] OR "multi-omics"[Title/Abstract] OR "multi
↪  omics"[Title/Abstract] OR multiomics[Title/Abstract] OR transcriptomics[Title/Abstract] OR
↪  genomics[Title/Abstract] OR proteomics[Title/Abstract] OR epigenomics[Title/Abstract] OR "cell
↪  type"[Title/Abstract] OR "gene regulatory"[Title/Abstract]) AND ("language model"[Title/Abstract] OR
↪  LLM[Title/Abstract] OR "large language model"[Title/Abstract] OR "natural language"[Title/Abstract] OR
↪  GPT[Title/Abstract] OR CLIP[Title/Abstract] OR "cross-modal"[Title/Abstract] OR multimodal[Title/Abstract] OR
↪  "multi-modal"[Title/Abstract] OR tokenization[Title/Abstract] OR "gene token"[Title/Abstract] OR
↪  prompt[Title/Abstract] OR "text generation"[Title/Abstract] OR "cell-to-text"[Title/Abstract] OR
↪  "text-to-cell"[Title/Abstract] OR NLP[Title/Abstract]) AND ("foundation model"[Title/Abstract] OR
↪  pretrained[Title/Abstract] OR "pre-trained"[Title/Abstract] OR "self-supervised"[Title/Abstract] OR
↪  transformer[Title/Abstract] OR generative[Title/Abstract] OR decoder[Title/Abstract] OR
↪  autoregressive[Title/Abstract] OR "transfer learning"[Title/Abstract] OR "zero-shot"[Title/Abstract] OR
↪  "few-shot"[Title/Abstract] OR "masked modeling"[Title/Abstract] OR "attention mechanism"[Title/Abstract] OR "deep
↪  learning"[Title/Abstract])) AND ("2026/07/07"[Date - Publication] : "2026/08/09"[Date - Publication]) AND "free
↪  full text"[sb] AND English[Language]
```

Scopus (executed 2026-08-10; 16 records retained)

```
TITLE-ABS-KEY(("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR
↪  scATAC-seq OR "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR
↪  "multi omics" OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene
↪  regulatory") AND ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR
↪  "cross-modal" OR multimodal OR "multi-modal" OR tokeniz* OR "gene token" OR prompt OR "text generation" OR
↪  "cell-to-text" OR "text-to-cell" OR NLP) AND ("foundation model" OR pretrain* OR pre-train* OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "masked modeling" OR "attention mechanism" OR "deep learning")) AND PUBYEAR > 2025 AND OPENACCESS(1) AND
↪  LANGUAGE(English)
```

OpenAlex (executed 2026-08-10; 100 records retained)

```
[main] (("single cell" OR "single-cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪  "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪  OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪  ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪  multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR "cell-to-text" OR
↪  "text-to-cell" OR NLP) AND ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer
↪  OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked
↪  modeling" OR "attention mechanism" OR "deep learning"))
[model_names] (scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR GeneGPT OR CellPLM OR Nicheformer
↪  OR EpiAgent OR PathOmCLIP)
```

Semantic Scholar (executed 2026-08-10; 183 records retained)

```
("single-cell" | "single cell" | "scRNA-seq" | "RNA-seq" | "RNA seq" | "gene expression" | transcriptomics | genomics
↪  | proteomics | epigenomics | "spatial transcriptomics" | "multi-omics" | "multi omics") + ("language model" | LLM
↪  | "large language model" | "natural language" | GPT | CLIP | multimodal | "multi-modal" | "cross-modal" | tokeniz*
↪  | prompt | NLP | "text generation") + ("foundation model" | pretrain* | "self-supervised" | transformer |
↪  generative | decoder | autoregressive | "transfer learning" | "zero-shot" | "deep learning")
```

arXiv (executed 2026-08-10; 14 records retained)

```
[primary] (ti:"single cell" OR ti:scRNA OR ti:"gene expression" OR ti:transcriptomics OR ti:genomics OR
↪  abs:"single-cell" OR abs:"RNA-seq" OR abs:"RNA seq" OR abs:"spatial transcriptomics" OR abs:"multi-omics" OR
↪  abs:"multi omics") AND (ti:"language model" OR ti:GPT OR ti:LLM OR ti:CLIP OR ti:multimodal OR abs:"language
↪  model" OR abs:"large language model" OR abs:"natural language" OR abs:"cross-modal" OR abs:"multi-modal" OR
↪  abs:tokeniz OR abs:prompt) AND (abs:"foundation model" OR abs:pretrain OR abs:"self-supervised" OR abs:transformer
↪  OR abs:generative OR abs:decoder OR abs:autoregressive OR abs:"transfer learning")
[cell_language] abs:"foundation model" AND abs:"single cell" AND abs:"language model"
[model_names] (ti:scGPT OR ti:tGPT OR ti:LangCell OR ti:ChatCell OR ti:GenePT OR ti:GeneGPT OR ti:CellPLM OR
↪  ti:Nicheformer OR ti:EpiAgent)
[bio_llm] abs:"gene expression" AND abs:LLM AND (abs:generative OR abs:decoder OR abs:autoregressive)
```

bioRxiv and medRxiv (Europe PMC preprints) (executed 2026-08-10; 20 records retained)

```
("single-cell" OR "single cell" OR "scRNA-seq" OR "RNA-seq" OR "RNA seq" OR "gene expression" OR "scATAC-seq" OR
↪  "ATAC-seq" OR "chromatin accessibility" OR "CITE-seq" OR "spatial transcriptomics" OR "multi-omics" OR "multi
↪  omics" OR transcriptomics OR genomics OR proteomics OR epigenomics) AND ("language model" OR LLM OR "large
↪  language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR multimodal OR "multi-modal" OR tokeniz*
↪  OR "gene token" OR prompt OR "text generation" OR NLP) AND ("foundation model" OR pretrain* OR "self-supervised"
↪  OR transformer OR generative OR decoder OR autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR
↪  "deep learning") AND (SRC:PPR) AND FIRST_PDATE:[2026-07-07 TO 2026-08-09]
```

Springer Nature (executed 2026-08-10; 27 records retained)

```
("single-cell" OR "single cell" OR scRNA-seq OR "RNA-seq" OR "RNA seq" OR "gene expression" OR scATAC-seq OR
↪  "ATAC-seq" OR "chromatin accessibility" OR CITE-seq OR "spatial transcriptomics" OR "multi-omics" OR "multi omics"
↪  OR multiomics OR transcriptomics OR genomics OR proteomics OR epigenomics OR "cell type" OR "gene regulatory") AND
↪  ("language model" OR LLM OR "large language model" OR "natural language" OR GPT OR CLIP OR "cross-modal" OR
↪  multimodal OR "multi-modal" OR tokenization OR "gene token" OR prompt OR "text generation" OR NLP) AND
↪  ("foundation model" OR pretrained OR "pre-trained" OR "self-supervised" OR transformer OR generative OR decoder OR
↪  autoregressive OR "transfer learning" OR "zero-shot" OR "few-shot" OR "masked modeling" OR "attention mechanism"
↪  OR "deep learning")
```

Google Scholar (executed 2026-08-10; 252 records retained)

```
("foundation model" "gene expression" "language model" single-cell) OR (GPT "single cell" "gene expression"
↪  generative) OR ("natural language" "single-cell" "cell type" foundation model) OR (multimodal LLM genomics
↪  transcriptomics "foundation model") OR (scGPT OR tGPT OR LangCell OR ChatCell OR CellWhisperer OR GenePT OR
↪  GeneGPT OR CellPLM OR Nicheformer OR EpiAgent) OR ("multi omics" "language model" "foundation model" transformer)
↪  OR ("RNA seq" LLM "foundation model" generative single-cell)
```

Date status of the retained records. PubMed: 27 within the window (database filter); Scopus: 16 within the window (exact-date check); OpenAlex: 100 within the window (database filter); Semantic Scholar: 144 within the window (exact-date check), 39 without a publication date (kept); arXiv: 14 within the window (database filter); bioRxiv and medRxiv (Europe PMC preprints): 20 within the window (database filter); Springer Nature: 27 within the window (database filter); Google Scholar: 252 with a publication year only (kept). Records removed as outside the window: 0.

Supplementary Note S2. Search: rounds, dates and source-specific details

The five rounds retained 5,534, 867, 933, 197 and 639 records (8,170).

Dates as executed. PubMed, arXiv, Europe PMC, Springer Nature and OpenAlex were queried with exact dates. Scopus and Semantic Scholar were queried by year, and records outside the window were then removed by exact date. Records removed (Scopus/Semantic Scholar): 18/1 in the baseline, then 165/226, 292/462, 389/766 and 485/914 in the top-ups. In round 5, 39 Semantic Scholar records without a publication date were kept. Google Scholar was queried by year only. The baseline ran on 2026-02-15, so no date-bounded query covered records dated 2026-02-16 to 2026-02-28.

Springer Nature. The free API searches full text and cannot be restricted to title and abstract. Retrieved records were therefore kept only if a term from each concept block appeared in the title or abstract. Raw to validated, by round: 17,130 to 272, 1,185 to 33, 1,129 to 27, 510 to 20 and 691 to 27. The flow counts the validated records.

Google Scholar completeness. The April and June rounds were rate-limited after 5 of 7 queries, and the July round returned no records. The baseline completion log was overwritten by a later Springer Nature-only rerun; the retained baseline records are preserved.

Gap between search windows. On 2026-10-14 the baseline queries of PubMed, arXiv, Europe PMC and Semantic Scholar were re-run for the window 2026-02-16 to 2026-02-28. They returned 51 unique records. 17 had already been retrieved and 34 had not; none of the 34 met the eligibility criteria in an indicative single-pass LLM title/abstract triage. Scopus and Springer Nature were not re-run.

Supplementary Note S3. Operation-level decomposition: rules, runs and stability

Library. Each documented input path is reconstructed as a directed graph of calls to a library of 22 reusable operations (normalisation, log transform, binning, tokenisation, table lookup, projection, concatenation, attention and others) with named ports and instance parameters, following the operation guide and the catalog definitions of release 1.3.0. Steps whose internal computation the paper does not state remain unexpanded boundaries. A four-paper pilot (ChatNT, InstructCell, X-Cell, MUPAD; 47 trajectories, 509 operation calls) was curated by the authors.

Agent decomposition. The remaining 50 papers were decomposed by agents (Codex CLI, gpt-5.6-luna, reasoning effort high; the earlier gpt-5.4-mini is no longer available through Codex) under version 3.2 of the decomposition protocol, which sharpens the boundaries between align, concatenate and serialize, and between partition and select, after run-to-run disagreements under version 3.1. The structure is fixed by the charted routes: one trajectory per charted configuration and one step per element of each route's verbatim transformation chain. For each step the agent returns only a decision, a closed category plus the library operation(s) named in the element text, with stated parameter values, verbatim evidence and the echoed chain element. A predefined rule set decides the category (for example unexpanded learned module, data provenance, external text generator, unspecified fusion). Routes without a recorded chain get a mechanical decision. Nodes, ports and data flow are compiled from these decisions, and the result is checked mechanically for verbatim evidence, catalog ports and declared parameters. The full paper text is supplied without truncation. Each paper was decomposed in three independent runs (up to four mechanical repairs each), and the published decision per step is the majority label. Steps without a majority remain unresolved boundaries and are listed for the library steward, as are steps decided 2 of 3.

Parameter pass. A separate parameter pass then read the whole paper again, with the operations fixed. For every declared parameter of every library call (2,433 slots in 1.3.0, defined in the operation parameter glossary) it decided whether the paper states the value for that data path, with verbatim evidence checked mechanically. Values that only follow from convention, released code or cited papers do not count. This pass also used three runs per paper and the majority status; its rules were tuned on the ten development papers (two iterations) and then frozen. Before the first run we fixed its gate: on ten held-out papers, consensus status agreement between runs 1-3 and 4-6 of at least 0.90. The result was library release 1.2.1. Under version 3.2, 39 of 1,468 steps changed operation; the parameter pass was repeated for the 16 affected papers (three runs), and slots of unchanged calls were carried over. The result is library release 1.3.0. No reviewer-agent pass was run on it.

Development and stability. The protocol was developed iteratively on ten length-stratified papers and evaluated on held-out papers with the rules frozen. Stability was measured as agreement between the consensus of runs 1-3 and of runs 4-6 against a predeclared gate. Of the 1,468 agent-decomposed steps, 1,394 (95.0%) were decided identically in all three runs and 74 (5.0%) by two of three; every step had a majority. Under version 3.1 with frozen rules, the consensus of runs 1–3 and of runs 4–6 on ten unseen papers agreed on 95.6% of steps and 85.4% of complete routes. After the boundary fixes of version 3.2, the same papers agree on 99.6% and 98.8%. That second figure is in-sample, because these papers informed the fixes. Across the three production runs under version 3.2 on all 50 agent-decomposed papers, 90.9% of route $\times$ run-pair comparisons had identical operation sequences (gate: at least 90%) and the median paper-level operation-set Jaccard was 1.0, but five papers fell below the predeclared minimum of 0.75 (lowest 0.0 for Chroma, where one run produced a disjoint set). The full-corpus gate was therefore not met; release decisions use the per-step majority of the three runs. For the parameter pass, 88% of slot decisions were unanimous across runs, and held-out consensus status agreement was 0.992. The withdrawn OKR-Cell preprint is not included.

Supplementary Note S4. Motifs, design axes and archetypes

Model-level paths. For each model configuration, the decomposed routes (agent-decomposed papers) or the author-curated graphs (pilot papers, all root-to-model paths) were turned into paths of role-aware operation tokens. Provenance and identity steps were dropped and repeated identical operations collapsed. Each operation's role follows from its inputs: biological only, text only, or joining biological and text inputs. Unexpanded learned modules were kept as typed tokens.

Motifs. Closed frequent sub-chains (contiguous and gapped) were mined at the operation and family levels, with support counted in papers (at least 4 papers; no length limit). Supported, interpretable sub-chains were named as motifs with templates, members, parameter reporting and evidence-quoted examples (Supplementary Table S6). Stability: mining was repeated on each single decomposition run.

Design axes. Each route was also coded on five axes along one canonical pipeline (source, curation, value transformation, carrier formation, interface): carrier formation, interface to the language model, transformation of measured values (order only, discretised, learned compression, verbatim; with subset and normalisation modifiers), text role and transparency. Coding rules are deterministic over the chains and the charted atlas labels (Supplementary Table S10). We tested coverage, independence (normalised mutual information) and robustness to decomposition runs. Archetypes are carrier x interface combinations; a model can carry several.

Earlier analysis. An earlier version of this analysis compared exact route chains and reported no recurring recipes. That result was an artefact of chain construction (uneven wording granularity, non-computational steps counted as operations, fragmentary routes) and is withdrawn.

Supplementary Note S5. Use of AI tools, reported by PRISMA-trAIce item

Tasks and tools. LLMs were run through the Codex command-line agent (Codex CLI; the version of the agent was not recorded with the runs). gpt-5.4-mini was used for title and abstract screening, full-text screening and route charting; gpt-5.6-luna with reasoning effort high was used for the operation-level decomposition, the parameter pass and the targeted charting pass, because the earlier gpt-5.4-mini is no longer available through Codex. Full texts were converted with Docling (version 2.110.0), and Docling Graph (version 1.7.0) reconstructed the complete data-source and input-representation sections that full-text screening used.

Screening. Title/abstract screening applied the eligibility decision tree with two role-separated LLM reviewers (scope and architecture) and an adjudicator for conflicts; prompts and schemas were hash-pinned. Full-text screening (baseline run on 2026-07-10) used Docling conversions; Docling Graph provenance reconstructed the complete data-source and input-representation sections, which were screened with the title and abstract. Records still uncertain after adjudication were resolved by the review lead from the full sections, keeping the automated decision alongside. The roles are prompts executed by the same model, not independent human reviewers.

Route charting. Routes were extracted from the full text with three frozen direct runs plus dense-coverage candidates. Agreement across the three runs on the 55-record corpus: route Jaccard 0.945–0.969; carrier-family exact agreement 0.929; Krippendorff's alpha 0.881 (acceptance thresholds 0.80, 0.90, 0.80). Dated author-led amendments are recorded in the atlas metadata.

Decomposition and parameter pass. Supplementary Note S3 gives the rules, runs and stability.

Targeted charting pass. The pass followed a frozen protocol (design-culture charting, version 1.1). Every item required verbatim evidence. Quotes were checked mechanically against the paper, with up to three repairs per run; all 108 runs passed validation. One quote broken by PDF line numbers was corrected by the steward. Run agreement was high for comparison presence (κ 0.92), LLM-generated text (κ 0.96) and dependence tests (Jaccard 0.91), but lower for evaluation formats (0.57) and rationales (0.60).

Adjudication and human involvement. The library steward is an agent-assisted role: an AI assistant, directed by the review lead, applies written rules to the verbatim evidence and records each decision with its reason. The steward adjudicated the rationale items, the single-run dependence tests and the cross-representation comparison units, and assigned each work to a culture. The charting runs returned 217 comparison items, which adjudication merged into 31 comparison units in 20 works; three further units were set aside as within-culture or not about representation form. A second human coder for the 31 comparison units and the culture assignment, with a recall audit of comparisons, is planned. The review lead approved the targeted charting pass and resolved the records left uncertain by automated full-text screening, and the authors curated the four pilot decompositions.

Prompts, schemas and run records. The prompts, output schemas, rule sets, run configurations and run outputs of every step are in the project repository; the supplementary file index lists their locations.

Supplementary Note S6. Coverage of protein and nucleotide models

A coverage check found that 12 protein and biomolecular text–generative models (ProteinDT, BioT5, BioT5+, NatureLM, Evolla, Galactica, InstructProtein, ProtLLM, Prot2Text, ProtT3, ProLLaMA, TxGemma) were never retrieved. None has a term of the biological-data block in its title or abstract. In the PubMed query for 2018-01-01 to 2026-08-09, a protein block in place of the biological-data block adds 1,140 records that the original block does not match (the original query returns 939), and a nucleotide block adds 508. In Europe PMC preprints the additions are 1,130 and 633 (hit counts on 2026-10-15).

Supplementary Tables

Supplementary Table S1. Deviations from the protocol

Amendments of the protocol after its first version under version control (version 3.0), and further changes that the protocol did not foresee. "Made after results were seen" records whether the change was decided after results of the review (or, where stated, after procedural results) were known.

No.	Kind	Date or stage	Area	Change	After results seen	Dependent results and labelling
1	Protocol amendment (version 3.1)	15 February 2026	Search	Space variants ('RNA seq', 'multi omics') were added to all queries and the search was re-run; this execution replaced the first one before screening.	No	All search results; reported as the baseline round.
2	Protocol amendment (version 3.2)	14 April 2026	Search and update	First top-up round across all sources; two models were added to the list of models that the search must find; short abstracts were enriched through Crossref title search.	No	Records of the April round.
3	Protocol amendment (version 4.0)	25 April 2026	Screening	Criterion-by-criterion screening with three prompt roles (scope reviewer, architecture reviewer, adjudicator) was adopted; the final decision is derived by fixed rules, not emitted by the model.	No	All screening decisions.
4	Protocol amendment (version 4.1)	10 June 2026	Search and update	Second top-up round with the same queries and only the date limits changed.	No	Records of the June round.
5	Protocol amendment (version 4.2)	6 July 2026	Search and update	Third top-up round with the same queries and only the date limits changed.	No	Records of the July round.
6	Protocol amendment (version 4.3)	10 July 2026	Screening	Full-text screening was done on the reconstructed data-source and input-representation sections of each paper, screened with the title and abstract.	No	Full-text screening decisions.
7	Protocol amendment (version 4.4)	10 July 2026	Screening	A human resolution layer was added: the review lead inspected the cited full sections of the records left uncertain by automated full-text screening; the automated decisions were kept alongside.	After the automated decisions	Inclusion decisions for the records left uncertain; the automated decisions are kept for audit.
8	Protocol amendment (version 4.5)	9 August 2026	Update procedure	A resumable procedure for living updates was added, from the dated search to publication of the updated atlas.	No	The fifth search round.
9	Protocol amendment (version 4.6)	10 August 2026	Search and update	Fifth round (living update): OpenAlex was added, and Google Scholar was queried with one combined query through a provider export.	No	Records of the August round.
10	Protocol amendment (version 4.7)	10 August 2026	Search and update	OpenAlex was added to abstract enrichment, so that no record of the August round is excluded for lack of an abstract.	No	Records of the August round.
11	Further change	After full-text screening	Data items	Input routes were charted as grounded routes from source object to carrier and classified in an input-representation taxonomy; the protocol's extraction plan did not foresee this.	No	All route-level results (Methods: data charting).
12	Further change	After route charting	Data items	Each input path was decomposed into calls to a library of reusable data operations, with a three-run consensus and a parameter pass that records whether each parameter is stated.	No	Operation reuse, unexpanded steps, unstated parameters, archetypes and opaque-step shares.
13	Further change	During the decomposition	Analysis	The decomposition rules were revised after run-to-run disagreement, sharpening the boundaries between align, concatenate and serialise, and between partition and select; the affected papers were decomposed and their parameter pass repeated.	Yes (stability results)	All decomposition results; the stability on the papers that informed the revision is reported as in-sample (Supplementary Note S3).
14	Further change	During the decomposition	Tool	The decomposition used gpt-5.6-luna instead of gpt-5.4-mini, which is no longer available through Codex.	No	Decomposition, parameter pass and targeted charting.

No.	Kind	Date or stage	Area	Change	After results seen	Dependent results and labelling
15	Further change	After the decomposition	Analysis	Motifs (recurring operation sub-chains) and five design axes with archetypes were defined after inspecting the decomposed data.	Yes	Motif and archetype results; the axes are described as not validated on an independent corpus (Limitations).
16	Further change	After the decomposition	Analysis	An earlier comparison of exact route chains, which reported no recurring recipes, was withdrawn as an artefact of chain construction and replaced by the motif analysis.	Yes	None; the earlier result is not reported.
17	Further change	After the decomposition	Analysis	The design-culture analysis and a targeted charting pass (rationales, supervision, dependence tests, cross-representation comparisons) were defined after exploring the decomposed corpus.	Yes	All culture-level results; described as exploratory (Methods, Results and figure legends), with contrasts not adjusted for multiple comparisons.
18	Further change	After screening	Correction	An error in the automated keyword check that precedes screening was corrected, which recovered one eligible record (XunZi); it was screened with the same frozen procedures.	Yes	Counted among the included records and shown in the flow diagram.
19	Further change	After the search	Search check	A 13-day gap between the baseline and first top-up windows was checked by re-running the baseline queries of four sources for the gap (2026-10-14); no eligible record was found.	Yes	None; reported in Limitations.
20	Further change	After the full-text screening	Screening check	An audit found that the abstract shown with the selected sections was cut for some records and that some reports were not screened at full text; the screening was not repeated.	Yes	None changed; reported in Limitations.
21	Further change	After the withdrawal of a preprint	Corpus	The withdrawn OKR-Cell preprint is still counted among the included records but was excluded from the decomposition and the culture-level analyses (author decision pending).	No	Record counts include it; work-level counts exclude it (Results).
22	Further change	At synthesis	Omission	Protocol objective 3 (datasets, downstream tasks, evaluation metrics) and the extraction fields for pretraining objective and code and data availability were not charted, except the evaluation formats and outputs of the targeted pass.	No	Not reported; listed in Limitations.

Supplementary Table S2. Search runs by round and source

Execution date, limits as executed, the number of hits reported by the source's interface and the records retained, for each round and source; "not searched" marks a source that was not searched in a round. The interface, the records fetched, the Springer Nature counts before validation and the export file of each run are in the data file.

Round	Source	Executed	Date limit as executed	Other limits as executed	Hits reported	Retained
1 baseline (protocol version 3.1)	PubMed	2026-02-15	[Date - Publication] 2018/01/01 to 2026/02/28	free full text[sb]; English[Language]	631	631
1 baseline (protocol version 3.1)	Scopus	2026-02-15	PUBYEAR > 2017; exact-date post-filter none to 2026-02-28	OPENACCESS(1); LANGUAGE(English)	1,039	1,021
1 baseline (protocol version 3.1)	OpenAlex	not searched				
1 baseline (protocol version 3.1)	Semantic Scholar	2026-02-15	year=2018-2026; exact-date post-filter none to 2026-02-28		2,193	2,192
1 baseline (protocol version 3.1)	arXiv	2026-02-15	submittedDate:[20180101 TO 20260228]	arXiv categories cs.LG/cs.AI/cs.CL/q-bio.GN/q-bio.QM/stat.ML	187	187

Round	Source	Executed	Date limit as executed	Other limits as executed	Hits reported	Retained
1 baseline (protocol version 3.1)	bioRxiv and medRxiv (Europe PMC preprints)	2026-02-15	FIRST_PDATE:[2018-01-01 TO 2026-02-28]	SRC:PPR (preprints)	669	669
1 baseline (protocol version 3.1)	Springer Nature	2026-02-15	datefrom:2018-01-01 dateto:2026-02-28	three-block title/abstract validation after retrieval		272
1 baseline (protocol version 3.1)	Google Scholar	2026-02-15	year range [2018, 2026] (year-level only)			562
2 April top-up	PubMed	2026-04-14	[Date - Publication] 2026/03/01 to 2026/04/14	free full text[sb]; English[Language]	46	46
2 April top-up	Scopus	2026-04-14	PUBYEAR > 2025; exact-date post-filter 2026-03-01 to 2026-04-14	OPENACCESS(1); LANGUAGE(English)	197	32
2 April top-up	OpenAlex	not searched				
2 April top-up	Semantic Scholar	2026-04-14	year=2026-2026; exact-date post-filter 2026-03-01 to 2026-04-14		427	201
2 April top-up	arXiv	2026-04-14	submittedDate:[20260301 TO 20260414]	arXiv categories cs.LG/cs.AI/cs.CL/q-bio.GN/q-bio.QM/stat.ML	14	14
2 April top-up	bioRxiv and medRxiv (Europe PMC preprints)	2026-04-14	FIRST_PDATE:[2026-03-01 TO 2026-04-14]	SRC:PPR (preprints)	5	5
2 April top-up	Springer Nature	2026-04-14	datefrom:2026-03-01 dateto:2026-04-14	three-block title/abstract validation after retrieval		33
2 April top-up	Google Scholar	2026-04-14	year range [2026, 2026] (year-level only)			536
3 June top-up	PubMed	2026-06-10	[Date - Publication] 2026/04/15 to 2026/06/10	free full text[sb]; English[Language]	68	68
3 June top-up	Scopus	2026-06-10	PUBYEAR > 2025; exact-date post-filter 2026-04-15 to 2026-06-10	OPENACCESS(1); LANGUAGE(English)	331	39
3 June top-up	OpenAlex	not searched				
3 June top-up	Semantic Scholar	2026-06-10	year=2026-2026; exact-date post-filter 2026-04-15 to 2026-06-10		677	215
3 June top-up	arXiv	2026-06-10	submittedDate:[20260415 TO 20260610]	arXiv categories cs.LG/cs.AI/cs.CL/q-bio.GN/q-bio.QM/stat.ML	21	21
3 June top-up	bioRxiv and medRxiv (Europe PMC preprints)	2026-06-10	FIRST_PDATE:[2026-04-15 TO 2026-06-10]	SRC:PPR (preprints)	33	33
3 June top-up	Springer Nature	2026-06-10	datefrom:2026-04-15 dateto:2026-06-10	three-block title/abstract validation after retrieval		27
3 June top-up	Google Scholar	2026-06-10	year range [2026, 2026] (year-level only)			530
4 July top-up	PubMed	2026-07-06	[Date - Publication] 2026/06/11 to 2026/07/06	free full text[sb]; English[Language]	25	25
4 July top-up	Scopus	2026-07-06	PUBYEAR > 2025; exact-date post-filter 2026-06-11 to 2026-07-06	OPENACCESS(1); LANGUAGE(English)	401	12
4 July top-up	OpenAlex	not searched				
4 July top-up	Semantic Scholar	2026-07-06	year=2026-2026; exact-date post-filter 2026-06-11 to 2026-07-06		884	118
4 July top-up	arXiv	2026-07-06	submittedDate:[20260611 TO 20260706]	arXiv categories cs.LG/cs.AI/cs.CL/q-bio.GN/q-bio.QM/stat.ML	6	6
4 July top-up	bioRxiv and medRxiv (Europe PMC preprints)	2026-07-06	FIRST_PDATE:[2026-06-11 TO 2026-07-06]	SRC:PPR (preprints)	16	16

Round	Source	Executed	Date limit as executed	Other limits as executed	Hits reported	Retained
4 July top-up	Springer Nature	2026-07-06	datefrom:2026-06-11 dateto:2026-07-06	three-block title/abstract validation after retrieval		20
4 July top-up	Google Scholar	2026-07-06	year range [2026, 2026] (year-level only)			0
5 August living update (protocol version 3.3)	PubMed	2026-08-10	[Date - Publication] 2026/07/07 to 2026/08/09	free full text[sb]; English[Language]	27	27
5 August living update (protocol version 3.3)	Scopus	2026-08-10	PUBYEAR > 2025; exact-date post-filter 2026-07-07 to 2026-08-09	OPENACCESS(1); LANGUAGE(English)	501	16
5 August living update (protocol version 3.3)	OpenAlex	2026-08-10	from_publication_ date:2026-07-07,to_ publication_date:2026-08-09	is_oa:true; language:en		100
5 August living update (protocol version 3.3)	Semantic Scholar	2026-08-10	year=2026-2026; exact-date post-filter 2026-07-07 to 2026-08-09		1,097	183
5 August living update (protocol version 3.3)	arXiv	2026-08-10	submittedDate:[20260707 TO 20260809]	arXiv categories cs.LG/cs.AI/cs.CL/q-bio.GN/q-bio.QM/stat.ML	14	14
5 August living update (protocol version 3.3)	bioRxiv and medRxiv (Europe PMC preprints)	2026-08-10	FIRST_PDATE:[2026-07-07 TO 2026-08-09]	SRC:PPR (preprints)	20	20
5 August living update (protocol version 3.3)	Springer Nature	2026-08-10	datefrom:2026-07-07 dateto:2026-08-09	three-block title/abstract validation after retrieval		27
5 August living update (protocol version 3.3)	Google Scholar	2026-08-10	year range [2026, 2026] (year-level only)			252

Supplementary Table S3. Reports excluded at full text, with reasons

One main reason per report, as recorded at full-text screening.

Reason	Reports
No natural-language text component	131
Application wrapper around an existing model	23
Not a generative model	13
Review or editorial	5
No biological data modality	3
Benchmark or resource without a new model	2
No foundation-model evidence	1
No substantive text-biology bridge	1
Total	179

Supplementary Table S4. Charted models

One row per charted model (109 models): the record, paper title, DOI and year, the number of routes and configurations, the input-carrier families, the primary carrier subtype, fusion topologies, text roles, lifecycle phases and biological modalities. Data file only.

Supplementary Table S5. Included works: design culture and charted variables

One row per independent work. Dependence test: whether the work reports any test of whether outputs depend on the biological input (found by both charting runs, or by one run and confirmed at adjudication). LLM-written text: whether text written by an LLM was used for training or evaluation. Language-model adaptation: how the language model is trained, where both charting runs agree (LoRA, low-rank adaptation; “none recorded” when no category was found by both runs). Cross-culture units: comparison units between arms of different cultures. Opaque steps: median share of the work’s biological route steps that are learned modules left unexpanded. Unstated slots: share

of the parameter slots of the proposed and component models whose value the paper does not state. A dash marks a work without a decomposed biological route or without parameter slots.

Study	Proposed models	Culture	Dependence test	LLM-written text	Language-model adaptation	Cross-culture units	Opaque steps	Unstated slots
Shen and Li 2024 [14]; Shen et al. 2026 [15]	OmniNA; OmniNA-1.7B	token-native	yes	no	full fine-tuning	0	0%	35%
Rizvi et al. 2025 [24]	C2S-Scale; C2S-Scale perturbation model	token-native	yes	yes	full fine-tuning	4	33%	39%
Fang et al. 2025 [27]	Cell-o1	token-native	yes	yes	LoRA or adapter	0	0%	50%
Gao et al. 2025 [28]	CellHermes	token-native	yes	yes	LoRA or adapter	0	0%	43%
Fang et al. 2024 [29]	CHATCELL	token-native	no	yes	none recorded	0	—	44%
Jiang et al. 2026 [31]	compound-phenotype scoring model; GEMGen	token-native	yes	yes	full fine-tuning	2	45%	41%
Tang et al. 2025 [32]	DeepSeek; DeepSeek-R1-Distill (1.5B); DeepSeek-R1-Distill 70B; DeepSeek-R1-Distill series (14B, 32B, and 70B)	token-native	no	yes	none recorded	1	—	89%
Liu et al. 2024 [33]	Gemma-7B; Geneverse; LLaMA2-13B; LLaMA2-7B; LLaMA3-8B; LLaMAPro-8B; LLaVA-7B (LoRA); Mistral-7B	token-native	no	yes	LoRA or adapter, full fine-tuning	0	0%	0%
Liang 2025 [34]	gene_eng_gpt2_para_seg; gene_eng_gpt2_summary; gpt2-gene-eng; gpt2-gene-eng-ft	token-native	no	no	full fine-tuning, trained from scratch	0	0%	19%
Shen et al. 2026 [35]	GenNA	token-native	yes	no	trained from scratch	0	50%	26%
Lyu et al. 2024 [37]	GP-GPT	token-native	no	no	LoRA or adapter	0	—	12%
Levine et al. 2023 [38]	GPT-2 Medium; GPT-2 Small	token-native	no	no	full fine-tuning, trained from scratch	0	0%	48%
Zhavoronkov et al. 2026 [42]	Longevity-LLM v0.1	token-native	no	yes	full fine-tuning	0	0%	100%
Li et al. 2025 [47]	Omni-DNA	token-native	no	yes	full fine-tuning	0	12%	7%
Wang 2026 [48]	OmniGene-4; OmniGene-4 v3	token-native	no	no	LoRA or adapter	1	0%	30%
Li et al. 2026 [52]	RVQ-Alpha	token-native	no	yes	full fine-tuning	0	0%	10%
Liu et al. 2025 [58]	TISSUENARRATOR	token-native	yes	no	LoRA or adapter	2	25%	4%
de Almeida et al. 2025 [5]	ChatNT	encoder-mediated	no	no	frozen	0	—	—
Fang et al. 2025 [6]	InstructCell	encoder-mediated	no	yes	full fine-tuning	1	—	—
Kharouiche et al. 2025 [10]; Markogiannakis 2026 [11]	Cell2Text-Gemma-4B; Cell2Text-Llama-1B; Cell2Text-Llama-1B-LoRA; Cell2Text	encoder-mediated	no	no	LoRA or adapter, full fine-tuning	0	50%	22%
Gupta et al. 2026 [19]	Bio-BLIP	encoder-mediated	yes	yes	frozen	1	17%	0%
Al-Kateb et al. 2025 [20]	BioGPT	encoder-mediated	yes	no	full fine-tuning	0	0%	57%
Luo et al. 2023 [21]	BioMedGPT-10B; BioMedGPT-LM-7B	encoder-mediated	no	no	frozen	2	67%	50%
Fallahpour et al. 2025 [22]	BIOREASON	encoder-mediated	no	yes	LoRA or adapter	1	40%	29%

Study	Proposed models	Culture	Dependence test	LLM-written text	Language-model adaptation	Cross-culture units	Opaque steps	Unstated slots
He et al. 2024 [25]	CaLMFlow	encoder-mediated	no	no	none recorded	0	0%	41%
Danner et al. 2025 [36]	Genolator	encoder-mediated	no	yes	full fine-tuning	1	50%	22%
Shi et al. 2025 [41]	scMMGPT	encoder-mediated	no	no	LoRA or adapter	1	25%	33%
Tu et al. 2023 [43]	Med-PaLM M	encoder-mediated	no	no	full fine-tuning	0	33%	17%
Sypetkowski et al. 2026 [46]	OmicsLM	encoder-mediated	no	yes	full fine-tuning	2	0%	19%
Xiang et al. 2026 [49]	OpticalDNA	encoder-mediated	yes	no	LoRA or adapter	1	40%	22%
Fesser et al. 2026 [51]	Qwen3-1.7B; Qwen3-1.7B and Qwen3-4B; Qwen3-1.7B and Qwen3-4B-Thinking; Qwen3-1.7B, Qwen3-4B, and Gemma 4 E2B	encoder-mediated	no	yes	LoRA or adapter	0	33%	14%
Xiao et al. 2026 [54]	SciCore-Omics	encoder-mediated	yes	yes	full fine-tuning	0	33%	58%
Wei et al. 2025 [56]	scMOBA	encoder-mediated	no	yes	LoRA or adapter	0	33%	43%
Schaefer et al. 2024 [12]; Schaefer et al. 2025 [13]	Mistral 7B	hybrid	yes	yes	full fine-tuning	2	50%	42%
Huang et al. 2026 [17]	XunZi-M; XunZi-R	hybrid	no	yes	full fine-tuning	1	50%	38%
Tsou et al. 2025 [23]	BIOVERSE	hybrid	no	yes	LoRA or adapter	2	33%	29%
Galkin et al. 2024 [26]	P3GPT	hybrid	yes	no	none recorded	0	17%	21%
Liu et al. 2025 [44]	TeamPath-7B	hybrid	no	yes	full fine-tuning	0	67%	70%
Zhang et al. 2026 [45]	OCellus; OCellus-Agent; OCellus-GNN	hybrid	yes	yes	LoRA or adapter	2	58%	34%
Queen et al. 2024 [50]	PROCYON; PROCYON-SPLIT	hybrid	yes	yes	none recorded	1	0%	79%
Chen et al. 2026 [59]	CellTosg2Sequence	hybrid	yes	no	LoRA or adapter	2	0%	29%
Wang et al. 2026 [7]	X-Cell; X-Cell-Ultra	language-conditioned	yes	no	full fine-tuning	1	—	—
Xiang et al. 2026 [8]	MUPAD	language-conditioned	no	no	none recorded	0	—	—
Sun et al. 2025 [18]	ALTER	language-conditioned	yes	no	none recorded	0	57%	27%
Ingraham et al. 2023 [30]	Chroma	language-conditioned	yes	no	none recorded	0	100%	0%
Gu et al. 2026 [39]	H2O	language-conditioned	yes	no	none recorded	0	100%	21%
Liu et al. 2025 [40]	LEONINE; LEONINE-GPT-2	language-conditioned	yes	no	full fine-tuning, trained from scratch	0	0%	17%
Tang et al. 2023 [53]	scDiff	language-conditioned	no	yes	none recorded	0	0%	14%
Zhang et al. 2025 [57]	Shusi	language-conditioned	yes	no	none recorded	0	58%	67%
Pang et al. 2025 [60]	TissueCraftAI	language-conditioned	no	no	none recorded	0	67%	67%
Shang et al. 2026 [55]	scMIR	alignment-only	no	no	none recorded	0	67%	36%

Supplementary Table S6. Motif library

Motifs are closed frequent sub-chains of role-aware operations supported by at least 4 papers. Operations: library operations in order. Input: whether the operations act on biological input, on text input, or join the two. Contiguous: whether the operations must be adjacent. Papers and models: support. The reporting of each motif's parameters and an example with its verbatim evidence are in the data file.

Motif	Family	Operations	Input	Contiguous	Papers	Models	Description
Subset then verbalise	carrier formation	select → serialise	biological input	no	10	17	Keep part of the biological record (top genes, fields, cells) and write it as text.
Ranked gene sentence	carrier formation	rank → serialise	biological input	no	5	12	Order features by value and write the ordered identifiers as text; magnitudes are dropped.
Expression normalisation	carrier formation	normalise → log transform	biological input	yes	5	7	Library-size or similar normalisation followed by a log transform of expression values.
Sample then verbalise	carrier formation	sample → serialise	biological input	no	4	5	Randomly draw cells, examples or negatives and write them into text.
Biological alphabet embedding	carrier formation	tokenise → table lookup	biological input	no	5	5	Tokenise a sequence in its own alphabet and look the tokens up in an embedding table.
Text token embedding	carrier formation	tokenise → table lookup	text input	no	6	7	Standard text channel: tokenise text and look the tokens up in the language model's embedding table.
Tile then encode	carrier formation	partition → learned module	biological input	no	7	8	Split an image, genome or sequence into patches/windows/chunks and encode them with a learned module.
Encoder then projector	encoding	learned module → learned projection	biological input	no	18	23	A learned (often pretrained, frozen) biological encoder followed by a projection into the receiving model's space.
Projector into the model	encoding	learned module → learned projection → learned module	biological input	no	6	8	Encoder output projected and then consumed by another learned module (typically the language model).
Stacked learned modules	encoding	learned module → learned module	biological input	no	12	14	Two or more unexpanded learned modules in sequence (e.g. frozen foundation model then trainable encoder).
Domain computation then encoder	encoding	computation outside the library → learned module	biological input	no	10	10	A computation outside the operation catalog (graph convolution, physics terms, clustering) before a learned encoder.
Verbalised biology read by a model	joining	serialise → learned module	biological input	no	9	17	Biology written as text and consumed by a learned language model.
Encode then join	joining	learned module → concatenate	biological input	no	6	6	Encoded biological representation concatenated with other inputs (text tokens, [CLS], other modalities).
Encode then add	joining	learned module → add	biological input	no	5	5	Encoded representation summed with another representation (positional, prior or other modality).
Text tokens into the model	joining	tokenise → learned module	text input	no	7	9	Tokenised text read by a learned module.

Supplementary Table S7. Gene and token budgets stated in the papers

Values as stated in the papers. (a) Genes or features kept per cell or object when a profile is written as text or passed to an encoder; "not stated" marks a route whose number the paper does not give. (b) Vectors, discrete codes or text tokens per object passed to the language model.

(a) Genes or features kept per cell or object.

Study	Model and setting	Route	Genes kept	Values kept	Evidence
Levine et al. 2023 [38]	Cell2Sentence (GPT-2)	text: proposed model input	100	no (rank order)	In order to limit computational costs, we truncate each cell sentence to only keep the 100 highest expressed (top 100 ranked) genes.
Rizvi et al. 2025 [24]	C2S-Scale	text: proposed model input	1000	no (rank order)	For predictive tasks such as cell type annotation on single cells, we find that K = 1000 often works well for cell annotation, achieving performance competitive or better than baseline models.
Gao et al. 2025 [28]	CellHermes	text: proposed model input	100	no (rank order)	Next, we ranked the gene expression values for each cell and truncated the top 100 genes for each cell , denoted as i .
Fang et al. 2024 [29]	ChatCell	text: proposed model input	100	no (rank order)	To capture the most significant expression patterns, we prioritize the top 100 most expressed genes, since lower-ranked genes tend to have more consistent expression values.
Liu et al. 2025 [44]	TeamPath	text: proposed model input	100	no (rank order)	To transfer the information in gene expression space to text space, we first rank the genes of each spot based on their expression profiles and select the top 100 genes to formulate them in natural language.
Chen et al. 2026 [59]	CellTosg2Sequence	text: proposed model input	200	no (rank order)	For a cell with normalised expression $X \in R\ G$ we sort genes by expression magnitude and keep the topK (K =200) HGNC symbols: Rank correlates almost linearly with log-expression ($R\ 2 \approx 0$.
Zhang et al. 2026 [45]	OCellus (cross-species task)	text: proposed model input	250	no (rank order)	We 5 probed cross-species reasoning with the cross-species cell-type task, in which OCellus is given a 6 mouse cell's top -ranked genes on the shared ortholog panel of approximately 250 genes and 7 asked to predict the canonical human-equivalent cell type.
Zhang et al. 2026 [45]	OCellus (EvenClock neighbours)	text: proposed model input	5	no (rank order)	Within each sector we sort neighboring cells by descending expression of their top-ranked genes and concatenate the top five marker genes per neighbor into a comma-separated string.
Schaefer et al. 2024 [12]	CellWhisperer (GPT-4/Mixtral conversation generation)	text: data-generation input	50	no (rank order)	For each transcriptome, we generated a chat-like conversation using one of two LLMs (GPT-4 or Mixtral) on the basis of the transcriptome's top 50 most highly expressed genes, the top 50 GSVA-derived gene sets, and the transcriptome's textual annotation.
Tsou et al. 2025 [23]	BIOVERSE baselines (open-domain LLMs)	text: comparator input	128	no (sorted by count)	### 5.2 MAIN RESULTS #### 5.2.1 ZERO-SHOT GENERATIVE CELL TYPE ANNOTATION We evaluate BIOVERSE's against two baselines on the PBMC10K dataset as discussed in scvi-tools (Gayoso et al., 2022) under zero-shot generation: (1) random and majority baseline (2) open-domain LLMs given a list of the 128 most expressed genes (sorted by expression count) as input.
Sypetkowski et al. 2026 [46]	OmicsLM text-only baselines	text: comparator input	50	no (rank order)	Text-only baselines receive the 50 highest-expressed genes per sample, with sample identifiers replaced by dummy labels (sample_1 , sample_2 , .
Sypetkowski et al. 2026 [46]	OmicsLM: C2S-Scale and Cell-o1 as comparators	text: comparator input	1000	no (rank order)	Candidate class labels are provided in text to OmicsLM, C2S-Scale-27B, and Cell-o1 during constrained likelihood scoring, matching the closed label set available to CLIP-like contrastive scoring; C2S-Scale and Cell-o1 receive the top-1000 expressed genes as text input.
Wang et al. 2026 [7]	X-Cell: C2S comparator	text: comparator input	1000	no (rank order)	We used the top 1,000 most highly expressed genes per cell (top k genes=1000), providing substantially more gene coverage than the 200-gene default in the C2S tutorial.
Schaefer et al. 2024 [12]	CellWhisperer (Geneformer encoder)	embedding: proposed model input	2048	encoder (rank tokens)	Specifically, the transcriptomes were sorted by gene expression levels, and the top 2048 most highly expressed genes were tokenized with a dictionary of human gene symbols (Theodoris et al.
Shi et al. 2025 [41]	scMMGPT (scGPT encoder)	embedding: proposed model input	2048	encoder (values)	To mitigate data sparsity, we retain the top 2,048 most expressed genes per cell, which captures the majority of meaningful signals.

Study	Model and setting	Route	Genes kept	Values kept	Evidence
Fang et al. 2025 [6]	InstructCell	embedding: proposed model input	3600	encoder (values)	To standardize gene features across all datasets, we identify the top 3,600 highly variable genes for each dataset's cell-by-gene matrix using Seurat [87, 88].
Jiang et al. 2026 [31]	GEMGen (signed DE lists)	text: proposed model input	50	no (rank order, sign as list)	Specifically, we selected the top 50/40/30 genes with the largest absolute CD values and then assigned each gene to the upregulated or downregulated category based on the sign of its value (positive or negative).
Fang et al. 2025 [27]	Cell-o1	text: proposed model input	not stated	no (rank order)	Each cell c i serves as a proxy for a cluster centroid and is represented by a ranked list of its top M expressed genes g i = [g i 1 , g i 2 , .
Sypetkowski et al. 2026 [46]	OmicsLM	embedding: proposed model input	20541	yes (vector)	For each sample, the default input vector concatenates four feature groups: a one-dimensional input-scale indicator, the normalized expression vector, a 512-dimensional Funomics T0 embedding [Powalski et al., 2026], and a 768-dimensional embedding from Geneformer-V2104M [Chen et al., 2026], yielding a 20,541-dimensional omics vector.
Xiang et al. 2026 [8]	MUPAD	embedding: proposed model input	331	yes (pathway scores)	Raw RNA-seq counts were normalized to TPM, filtered to remove housekeeping and mitochondrial genes, and compressed into 331 pathway-level enrichment scores using established pan-cancer gene signatures 61 , yielding biologically interpretable conditioning inputs.
Wang et al. 2026 [7]	X-Cell	embedding: proposed model input	4000	yes (values)	| Parameter | Mini (60M) | X-Cell-Ultra Phase 1 | X-Cell Ultra Phase 2 | |———————-|———————|———————|———————| | Dataset | Replogle-Nadig | Top 5% high-effect | 100% all perturbations | | Nodes (GPUs) | 1 (8) | 12 (96) | 16 (128) | | Set size S | 64 | 64 | 64 | | Optimizer step size | 64 | 96 | 128 | | Effective batch (cells) | 4,096 | 6,144 | 8,192 | | Epochs | 6 | 10 | 10 | | Optimizer | AdamW | AdamW | AdamW (fresh) | | β 1 ,β 2 | 0.9, 0.999 | 0.9, 0.999 | 0.9, 0.999 | | Peak LR | 10 - 4 | 3 × 10 - 4 | 4 × 10 - 5 | | Weight decay | 0.01 | 0 | 0 | | LR schedule | Cosine + warmup | Cosine + warmup | Cosine + warmup | | Warmup | 10% | 10% | 3% | | Grad clip norm | 1.0 | 1.0 | 1.0 | | Context length G ′ | Full (∼ 19k) | 4,000 | 4,000 | | Mixed precision | bf16 | bf16 | bf16 | | FSDP strategy | N/A | Per-layer | Per-layer | | Warmstart | scGPT embeddings | - | Phase 1 epoch 9 | #### Batch size scaling.

(b) Vectors, codes or text tokens per object.

Study	Model	What is counted	Number	Evidence
Fang et al. 2025 [6]	InstructCell	Q-Former query tokens per cell	8	For the Q-Former module, L cell is set to 4, the number of query tokens k is 8, and the number of key-value pairs t is 6.
Shi et al. 2025 [41]	scMMGPT	Q-Former queries per cell	32	- The cell-to-text projector is implemented with a Query Transformer (Q-Former) [33] with 32 learnable queries, which maps high-dimensional cell embeddings from the scLLM into the token embedding space of the Text LLM.
Danner et al. 2025 [36]	Genolator	virtual tokens per modality embedding	8	For model input, each embedding modality was mapped to 8 virtual tokens of size 4096.
Gupta et al. 2026 [19]	Bio-BLIP	Q-Former query vectors (AlphaGenome / GenePT / ESM)	32	The Alphagenome Q-former has 32 query vectors, while the GenePT Q-former and ESM Q-former are trained with 8 query vectors each.
Shang et al. 2026 [55]	scMIR	Q-Former query embeddings per image	32	In our implementation, we use 32 learnable query embeddings, each with a hidden dimension of 768.

Study	Model	What is counted	Number	Evidence
Li et al. 2026 [52]	RVQ-Alpha	discrete tokens per cell	10	We present RVQ-Alpha , an end-to-end framework that bridges this modality gap through three contributions: (i) a Residual Vector Quantization (RVQ) tokenizer that compresses each cell into a fixed 10-token sequence via eight residual codebooks embedded directly in the LLM vocabulary, requiring 3.4 × fewer tokens than prior discrete methods and enabling bidirectional unification of cell interpretation and generation; (ii) scCoT-Synth , a teacher-student engine that grounds newly added biological tokens through evidence-before-conclusion reasoning, using the language modeling objective as the cross-modal alignment signal without a separate projection network; and (iii) a Fact-Aware RLVR system combining an ontology-grounded answer judge with saliency-weighted verification of biological claims against actual expression data, under dynamic gating that conditions hallucination suppression on task competence.
Chen et al. 2026 [59]	CellTosg2Sequence	KG virtual tokens per cell (one per gene)	200	Each prompt has four blocks: (1) KG virtual tokens [KG_VT] : one virtual token per input gene (block length K =200), aligned one-to-one with the Cell Sentence; (2) Biological metadata [Meta] : short natural-language fields (tissue, compartment, disease, optional KG triples); (3) Cell Sentence [CellSentence] : the ranked gene list as HGNC symbols; (4) Task instruction [Task] : one short line specifying the output format.
Tsou et al. 2025 [23]	BIOVERSE	[BIO] embeddings per object (notation k=1)	1	Given a query q , bio embeddings ˜ z b injected at a 1 For notational simplicity, we assume k = 1 and omit the token index in most equations from here onward; the framework naturally extends to k > 1 when multiple embeddings are injected [BIO] marker, and paired target text t b = (t 1 , .
Levine et al. 2023 [38]	Cell2Sentence	text tokens: prompt + top-100 genes	350	On average, the number of tokens in the prompts and top 100 genes combined was around 350.
Li et al. 2026 [52]	RVQ-Alpha (critique of ranked lists)	text tokens for top-20 gene list	80	However, these approaches introduce a different fundamental trade-off: ranked gene lists discard quantitative expression precision and produce token-inefficient representations (40-80 tokens per cell for the top-20 genes), whereas structured descriptions scale poorly to population-level multi-cell reasoning [28].
Wei et al. 2025 [56]	scMOBA	cross-attention learnable queries	256	To effectively summarize informative biological signals, 256 learnable queries are introduced to perform cross-attention with the GCN outputs, extracting the most relevant features for downstream textual reasoning while reducing the input length to the language model.
Xiao et al. 2026 [54]	SciCore-Omics	Q-Former query virtual tokens per profile	32	In the implementation used in this study, the gene branch follows where the gene encoder first maps the gene expression profile into molecular embeddings, the Q-Former compresses the molecular representation into 32 query virtual tokens, and the projector maps these virtual tokens into the language-model hidden space.
Sypetkowski et al. 2026 [46]	OmicsLM	projected vector per placeholder (one per profile)	1	Bottom: each placeholder is built by mapping a single-cell or bulk expression profile, including profiles from perturbation screens, to a continuous vector and projecting it into the LLM token-embedding space, allowing multiple profiles to be interleaved in one prompt for sample-level and comparative reasoning.

Supplementary Table S8. Text written by LLMs and used for training or evaluation

Works that train or evaluate on text written or rewritten by another LLM. What was written: training or evaluation content, or instruction templates only. Human check: whether the paper reports a human check of the generated text. Reasoning traces: whether the generated text includes reasoning traces. Used in evaluation: whether LLM-written text enters the evaluation.

Study	Culture	What was written	Generators	Human check	Reasoning traces	Used in evaluation	Example
Rizvi et al. 2025 [24]	token-native	training or evaluation content	GPT-4o; GPT-4.5; GPT-3.5-Turbo	no	no	no	To create training data for this task, we use GPT-4o to generate natural language captions for cell clusters derived from annotated datasets (Methods Section 4.7).
Fang et al. 2025 [27]	token-native	training or evaluation content	OpenAI o1	no	yes	no	We begin by performing reasoning distillation using OpenAI's o1 [22], a frontier LLM with strong multi-step reasoning capabilities, to construct a synthetic dataset of high-quality reasoning traces and predictions.
Gao et al. 2025 [28]	token-native	instruction templates only	ChatGPT-4o	no	no	no	To increase linguistic diversity and robustness, we used ChatGPT-4o 39 to augment instruction types with 20 diverse task descriptions 48 (Methods).
Fang et al. 2024 [29]	token-native	instruction templates only	GPT-3.5-turbo	no	no	no	This description is then fed into GPT-3.5turbo, leveraging its capability to produce diverse renditions of the same concept.
Jiang et al. 2026 [31]	token-native	training or evaluation content	GPT-4o	yes	no	no	We curated descriptive annotations for all cell types used in this study by combining two sources: descriptions retrieved from the ATCC cell catalog, and independently generated summaries by GPT-4o, which were manually reviewed to ensure biological plausibility.
Tang et al. 2025 [32]	token-native	training or evaluation content	GPT-OSS-120B	no	no	no	We use the NCBI-derived biological dataset from GenePT [26]: biomedical corpus from NCBI is used as the knowledge source in EasyDataset [30], and GPT-OSS-120B [31] synthesizes instruction-response pairs covering broad biological knowledge.
Liu et al. 2024 [33]	token-native	training or evaluation content	GPT-3.5	no	no	no	To generate LLMs for genomic and proteomic analysis, we incorporate the training datasets from rephrased descriptions for gene functions as well as synthetic descriptions from GPT 3.5.
Zhavoronkov et al. 2026 [42]	token-native	unclear	generator not stated	no	no	no	To improve Longevity-LLM's generalization ability, prompt collections feature semantic variation, formatting options, synonym rotation, and automatically generated reasoning traces.
Li et al. 2025 [47]	token-native	training or evaluation content	LLMAnnotator (model not named)	no	no	no	A LLMAnnotator is employed to extend and refine functional annotations.

Study	Culture	What was written	Generators	Human check	Reasoning traces	Used in evaluation	Example
Li et al. 2026 [52]	token-native	training or evaluation content	unspecified teacher LLM (replaceable backend)	no	yes	no	We present RVQ-Alpha , an end-to-end framework that bridges this modality gap through three contributions: (i) a Residual Vector Quantization (RVQ) tokenizer that compresses each cell into a fixed 10-token sequence via eight residual codebooks embedded directly in the LLM vocabulary, requiring 3.4 × fewer tokens than prior discrete methods and enabling bidirectional unification of cell interpretation and generation; (ii) scCoT-Synth , a teacher-student engine that grounds newly added biological tokens through evidence-before-conclusion reasoning, using the language modeling objective as the cross-modal alignment signal without a separate projection network; and (iii) a Fact-Aware RLVR system combining an ontology-grounded answer judge with saliency-weighted verification of biological claims against actual expression data, under dynamic gating that conditions hallucination suppression on task competence.
Fang et al. 2025 [6]	encoder-mediated	instruction templates only	GPT-4o	no	no	no	Next, we prompt GPT-4o to generate a set number of instruction-response templates for each task.
Gupta et al. 2026 [19]	encoder-mediated	training or evaluation content	Claude Sonnet 4.6	no	yes	no	We construct 5-step chain-of-thought reasoning traces linking GWAS variants from the Gold Standard Dataset to target genes and to the observed phenotype using Claude Sonnet 4.6 and ground-truth expert annotations from the available under a CC-BY 4.0 International license.
Fallahpour et al. 2025 [22]	encoder-mediated	training or evaluation content	Claude 3.7 Sonnet; GPT-4o	no	yes	no	These sequences have an average length of approximately 4,000 base pairs, with most variants differing by only 1-3 nucleotides from their reference sequences. #### 4.1.2 Reasoning Path Construction and Curation Adistinctive feature of this dataset is its inclusion of explicit causal reasoning paths connecting genetic variants to disease phenotypes via defined molecular mechanisms; these paths were constructed using the Claude 3.7 Sonnet model [2] and grounded with contextual disease information from the KEGG disease database [20].
Danner et al. 2025 [36]	encoder-mediated	training or evaluation content	GPT-4.1	yes	no	yes	For each GO aspect 29 , we engineered specific system prompts to guide a GPT-4.1 37 model (hosted on Azure AI Foundry) in generating three confirmation QA samples per gene, leveraging the ontology annotations and at least six manually crafted few-shot examples from two or more unrelated genes per prompt.
Sypetkowski et al. 2026 [46]	encoder-mediated	training or evaluation content	unnamed LLM curation pipeline	no	no	no	We addressed this with an LLM-based curation pipeline that (i) generated free-text natural language descriptions for each sample, and (ii) extracted structured annotations including tissue, cell type, cell line, disease, and perturbation target.

Study	Culture	What was written	Generators	Human check	Reasoning traces	Used in evaluation	Example
Fesser et al. 2026 [51]	encoder-mediated	training or evaluation content	Claude 3.7 Sonnet (inherited BIOREASON traces)	no	yes	yes	In BIOREASON, these traces were produced using Claude 3.7 Sonnet and grounded with contextual disease information from the KEGG disease database [78, 79], then packaged into standardized question-answer pairs for training and evaluation [1].
Xiao et al. 2026 [54]	encoder-mediated	training or evaluation content	unspecified LLM	no	no	no	Panels: - **a**: Schematic from a gene expression matrix to extracted biological features: top expressed/marker genes, pathway enrichment via ssGSEA, and tissue/metadata, which are converted into LLM-generated text.
Wei et al. 2025 [56]	encoder-mediated	training or evaluation content	unspecified LLM (LLM-assisted manual curation)	yes	no	no	To ensure the consistency of metadata from the large-scale and diverse high-dimensional datasets, we applied an LLMassisted manual curation procedure to align the cell type taxonomy, brain region origins, developmental stages and disease status from different datasets (Extended Data Fig. 1c and Supplementary Table 2).
Schaefer et al. 2024 [12]; Schaefer et al. 2025 [13]	hybrid	training or evaluation content	GPT-4; Mixtral 8x7b; GPT-4o	no	no	yes	Our method was created in three steps (Fig. 1a): (1) LLM-assisted curation of multimodal training data, resulting in 1,082,413 pairs of human RNA-seq profiles and matched textual annotations; (2) training of the CellWhisperer embedding model, which places the transcriptomes and their AI-curated textual descriptions into a joint embedding space for cell search and annotation; and (3) development of the CellWhisperer chat model for transcriptome-aware question answering and natural-language chats.
Huang et al. 2026 [17]	hybrid	training or evaluation content	GPT-4	yes	yes	no	These entries were generated using structured prompts with GPT-4 to transform raw gene-disease associations from databases, including DisGeNET 10 and the Comparative Toxicogenomics Database (CTD) 13 , into detailed mechanistic interpretations based on literature evidence.
Tsou et al. 2025 [23]	hybrid	training or evaluation content	unspecified LLM (inherited dataset descriptions)	no	no	no	Following the dataset protocol, we use the CellxGene subset (Perkel, 2024), where pseudo-bulk RNA samples are generated by averaging single-cell profiles, and natural-language descriptions are produced from cell and tissue metadata using large language models.
Liu et al. 2025 [44]	hybrid	training or evaluation content	o4-mini	yes	yes	no	Reasoning data were constructed using COT templates generated based 94 on the advanced reasoning model o4-mini (25), with subsequent quality validation performed by 95 pathologists at Yale School of Medicine.
Zhang et al. 2026 [45]	hybrid	instruction templates only	GLM-5.2	no	no	no	The Planner LoRA is fine-tuned on five thousand natural-language to directedacyclic-graph pairs synthesized via an Anthropic-compatible GLM-5.2 applicationprogramming-interface endpoint

Study	Culture	What was written	Generators	Human check	Reasoning traces	Used in evaluation	Example
Queen et al. 2024 [50]	hybrid	training or evaluation content	OpenAI GPT models	no	no	no	To increase the diversity of phenotypes, we rephrase the original descriptions using OpenAI's GPT models.
Tang et al. 2023 [53]	language-conditioned	training or evaluation content	GPT-4	no	no	yes	For these two terms, we utilize GPT4 (OpenAI, 2023) to depict their descriptions given available definitions as query contexts.

Supplementary Table S9. Cross-culture comparison units

One row per comparison unit after adjudication: one comparison between arms of different cultures, per work and task family. Design: a controlled ablation within one system, the same backbone given the data as text, or different systems. Opponent arm: its culture and how it was obtained. Values reported: as quoted in the charting; a dash marks a unit without values in the text. Image rendering is the DNA-as-image arm of OpticalDNA.

Study	Task family	Task	Design	Proposed arm	Opponent arm (obtained as)	Winner	Values reported
Liu et al. 2025 [58]	spatial	spatial coordinates: numeric tokens vs learned positional embeddings	controlled ablation within one system	token-native	encoder-mediated (trained ablation)	token-native	qualitative only
Liu et al. 2025 [58]	quantitative generation	cell generation / ranked gene prediction vs Nicheformer	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	NDCG, overlap (values not in text)
Queen et al. 2024 [50]	text generation	phenotype generation vs LLMs given gene symbol/UniProt ID/sequence	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	BERTScore F1; LLM judge
Rizvi et al. 2025 [24]	text generation	cluster captioning vs CellWhisperer	different systems	token-native	encoder-mediated (external pretrained model)	token-native	BioBERTScore
Rizvi et al. 2025 [24]	categorical	cell type annotation vs expression-only FMs	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	>85% accuracy
Rizvi et al. 2025 [24]	embedding quality	cell embedding linear probe vs expression FMs	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	—
Rizvi et al. 2025 [24]	categorical	spatial niche prediction vs averaged embeddings	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	—
Schaefer et al. 2024 [12]; Schaefer et al. 2025 [13]	text generation	chat answers: embedding + top-50 gene list vs embedding only	controlled ablation within one system	hybrid	encoder-mediated (trained ablation)	hybrid	mild benefit (perplexity)
Schaefer et al. 2024 [12]; Schaefer et al. 2025 [13]	text generation	chat answers vs Mistral 7B / Llama 3.3 70B given top-50 genes	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	lowest perplexity
Fallahpour et al. 2025 [22]	sequence-level	KEGG reasoning, variant effect, chromatin: DNA encoder + Qwen3 vs Qwen3 reading DNA as text	same backbone given the data as text	encoder-mediated	token-native (trained baseline)	encoder-mediated	VEP coding acc 80.2 vs 49.0
Fang et al. 2025 [6]	quantitative generation	pseudo-cell generation and drug sensitivity vs Cell2Sentence	different systems	encoder-mediated	token-native (trained baseline)	encoder-mediated	—
Shi et al. 2025 [41]	categorical	cell type annotation: expression values vs cell sentence	controlled ablation within one system	encoder-mediated	token-native (trained ablation)	encoder-mediated	error 0.76% vs 6.42%
Xiang et al. 2026 [49]	sequence-level	eQTL: DNA rendered as image vs 1D tokens at matched budget	controlled ablation within one system	image rendering	token-native (trained ablation)	image rendering	AUROC 0.626 vs 0.558

Study	Task family	Task	Design	Proposed arm	Opponent arm (obtained as)	Winner	Values reported
Tsou et al. 2025 [23]	categorical	zero-shot PBMC annotation vs LLMs given 128-gene list	different systems	hybrid	token-native (prompted general LLM)	mixed	acc 0.614 vs GPT-OSS-120B 0.779, Granite-8B 0.369
Tsou et al. 2025 [23]	text generation	molecule/protein description vs LLMs given raw SMILES/sequence	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	LLM-judge 0.17 vs 0.02
Danner et al. 2025 [36]	categorical	GO confirmation/denial vs GPT-4.1 given sequences as text	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	MCC 0.955 vs 0.18
Tang et al. 2025 [32]	embedding quality	LLM-written gene profiles + text embedding vs scGPT	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	gene-level AUC mostly higher
Luo et al. 2023 [21]	text generation	protein QA: aligned encoder features vs raw sequence into the same Llama2-7B-Chat	controlled ablation within one system	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	BLEU-2 0.344 vs 0.019
Luo et al. 2023 [21]	text generation	molecule QA vs ChatGPT/Llama2 given SMILES	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	BLEU-2 0.234 vs 0.103
Wang et al. 2026 [7]	quantitative generation	perturbation response vs C2S-2B cell sentences	different systems	encoder-mediated	token-native (trained baseline)	encoder-mediated	Pearson delta 0.51 vs -0.03 (iPSC-200)
Jiang et al. 2026 [31]	embedding quality	cross-dataset consistency: ranked DEG text vs numeric DE vectors	different systems	token-native	encoder-mediated (representation compared without a model)	token-native	—
Jiang et al. 2026 [31]	other generation	molecule generation vs GexMolGen (expression profiles)	different systems	token-native	encoder-mediated (trained baseline)	token-native	hit recovery
Chen et al. 2026 [59]	categorical	cell type / drug response: KG virtual tokens + cell sentence vs cell sentence only	controlled ablation within one system	hybrid	token-native (trained ablation)	hybrid	0.385 vs 0.268; 0.606 vs 0.487
Chen et al. 2026 [59]	categorical	immune cell type annotation vs C2S-Scale-27B	different systems	hybrid	token-native (external pretrained model)	token-native	exact 24.1% vs 85.6%
Gupta et al. 2026 [19]	sequence-level	variant annotation / target gene vs LLM given AlphaGenome outputs or variant context as text	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	target gene 71.2% vs 64.4%
Sypetkowski et al. 2026 [46]	multi-task	10-task suite: embedding stack vs Cell2Sentence-like top-100 text	controlled ablation within one system	encoder-mediated	token-native (trained ablation)	encoder-mediated	relative rank .740 vs .090
Sypetkowski et al. 2026 [46]	categorical	GEO-OmicsQA vs text-only LLMs given 50 genes (incl. C2S-Scale-27B, Cell-o1, GPT-5.5)	different systems	encoder-mediated	token-native (prompted general LLM)	encoder-mediated	binary QA 0.752 vs <=0.588
Wang 2026 [48]	categorical	remote homology: biological tokens vs ESM-2 embeddings	different systems	token-native	encoder-mediated (external frozen embedding)	token-native	59.5% vs 50.5%
Huang et al. 2026 [17]	categorical	regulator prediction: combined vs text-only XunZi-R and graph-only XunZi-M	different systems	hybrid	token-native (trained baseline)	hybrid	AUC 0.92 vs GPT-4o 0.52
Zhang et al. 2026 [45]	embedding quality	cell embeddings vs 13 single-cell FMs	different systems	hybrid	encoder-mediated (external frozen embedding)	mixed	linear probe 95.1% (best) but NMI below scimilarity

Zhang et al. 2026 [45]	quantitative generation	perturbation response: LLM-derived gene embeddings vs GO features (GEARS)	different systems	hybrid	encoder-mediated (external frozen embedding)	hybrid	Pearson 0.945 vs 0.84

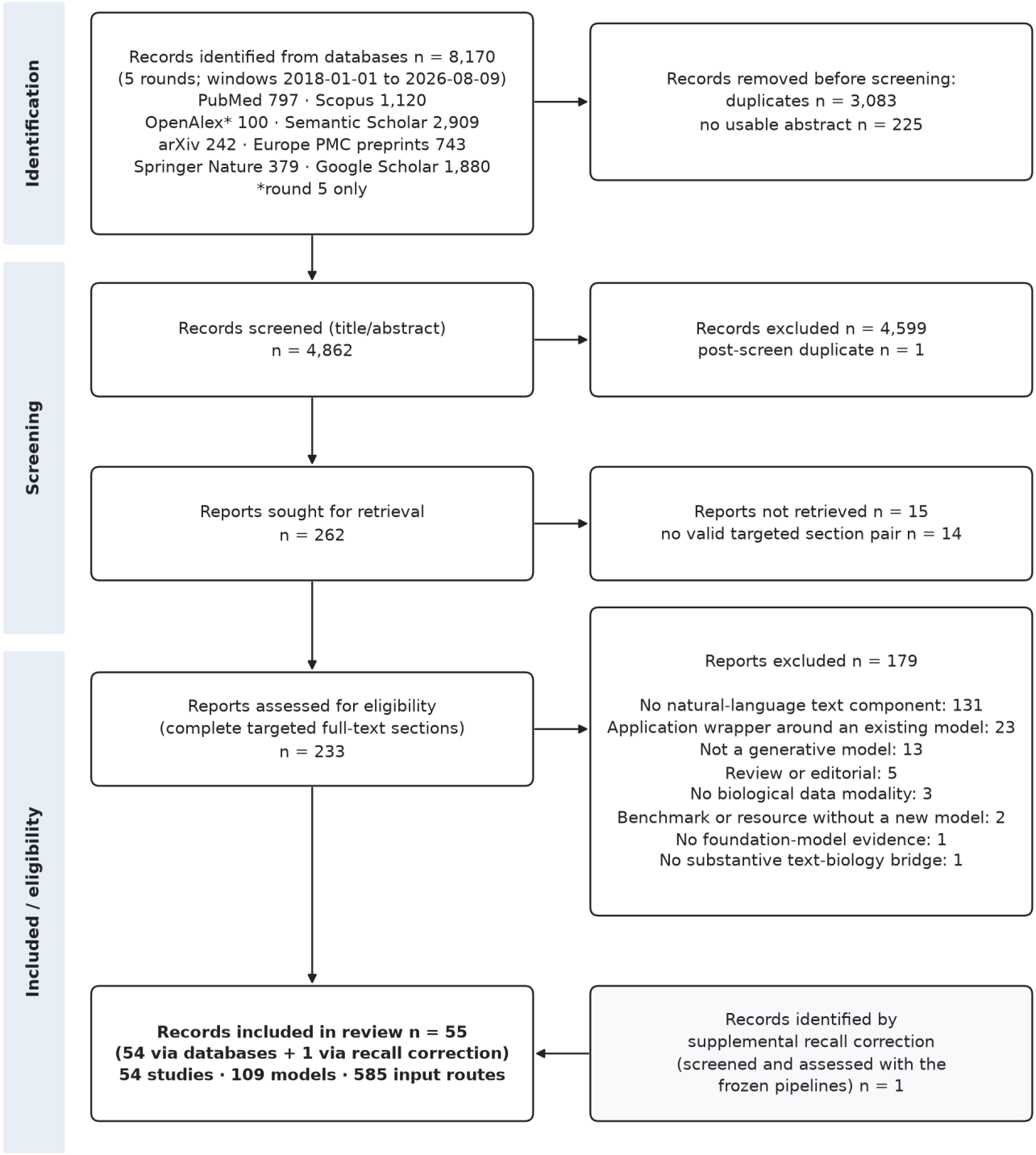
Supplementary Table S10. Route-level design-pattern codebook

One row per charted route of the 50 agent-decomposed papers (532 routes): the model and configuration, lifecycle phase, source modality, carrier family and subtype, fusion topology, text role, the operation chain, the value of each design axis with the rule that assigned it, the share of opaque steps, the share of stated parameters and the archetype. Data file only.

Supplementary Figures

Supplementary Figure S1. Full PRISMA 2020 flow, with the records identified per database

Records identified per source and round, records removed before screening, records screened and excluded, reports sought, not retrieved and assessed, and studies included. Reasons for exclusion at full text are given in Supplementary Table S3.



Data files and column dictionary

The data files are comma-separated, UTF-8 encoded and use plain-language values. The coded originals from which they are built are kept in the project repository (see Data and code availability).

File	Item	Contents	Rows
deviations.csv	Supplementary Table S1	Deviations from the protocol	22
search_runs.csv	Supplementary Table S2	Search runs by round and source, with limits and counts as executed	40
fulltext_exclusions.csv	Supplementary Table S3	Reports excluded at full text, by reason	8
charted_models.csv	Supplementary Table S4	Charted models with their records, routes and input labels	109
included_works.csv	Supplementary Table S5	Included works with their design culture and charted variables	51
motif_library.csv	Supplementary Table S6	Motif library: recurring operation sub-chains with an example each	15
gene_budgets.csv	Supplementary Table S7	Genes or features kept per cell or object, as stated in the papers	21
token_budgets.csv	Supplementary Table S7	Vectors, codes or text tokens per object passed to the language model	13
llm_written_text.csv	Supplementary Table S8	Works that used text written by LLMs for training or evaluation	25
comparison_units.csv	Supplementary Table S9	Cross-culture comparison units after adjudication	31
route_codebook.csv	Supplementary Table S10	Route-level codebook of the design axes and archetypes	532

Column dictionary (also provided as data_dictionary.csv).

File	Column	Description
deviations.csv	number	Number of the deviation
deviations.csv	kind	Protocol amendment (with the version it created) or further change not foreseen by the protocol
deviations.csv	date_or_stage	Date of the amendment or the stage at which the change was made
deviations.csv	area	Part of the review affected
deviations.csv	change	What changed
deviations.csv	made_after_results_were_seen	Whether the change was decided after results were known
deviations.csv	dependent_results_and_labelling	Results that depend on the change, and how they are labelled
search_runs.csv	round	Search round: number, name and the protocol version under which it ran
search_runs.csv	window_start	First publication date of the round's search window (ISO date)
search_runs.csv	window_end	Last publication date of the round's search window (ISO date)
search_runs.csv	source	Bibliographic source
search_runs.csv	searched	Whether the source was searched in this round (yes or no)
search_runs.csv	executed_on	Date on which the search was executed (ISO date)
search_runs.csv	interface	Interface through which the source was queried
search_runs.csv	date_limit_as_executed	Date restriction as executed, in the syntax of the source
search_runs.csv	other_limits_as_executed	Other limits as executed (language, access, categories, validation)
search_runs.csv	hits_reported	Total number of results reported in the export metadata, where the interface reports one
search_runs.csv	records_fetched	Number of records fetched, as recorded in the export metadata
search_runs.csv	records_before_validation	Records in the raw export before filtering after retrieval (for Springer Nature, the check that a term of each concept block is in the title or abstract)
search_runs.csv	count_in_run_log	Count for the source in the search summary file of the round
search_runs.csv	records_retained	Records in the export file of the run, passed to deduplication
search_runs.csv	export_file	Repository path of the export file of the run
fulltext_exclusions.csv	reason	Main reason for exclusion at full text
fulltext_exclusions.csv	reports	Number of reports excluded for this reason
charted_models.csv	model	Model name as charted
charted_models.csv	study	First author and year of the record, as in the article's reference list
charted_models.csv	record_id	Identifier of the included record in the review database
charted_models.csv	paper_title	Title of the record
charted_models.csv	doi	DOI of the record
charted_models.csv	year	Publication year
charted_models.csv	year_basis	Source of the publication year
charted_models.csv	routes	Number of charted input routes of the model
charted_models.csv	configurations	Number of charted configurations (training or use settings) of the model
charted_models.csv	input_carrier_families	Families of the carriers through which inputs reach the model
charted_models.csv	primary_carrier_subtype	Most frequent carrier subtype among the model's routes

File	Column	Description
charted_models.csv	fusion_topologies	How the inputs are fused with the language model, over all routes
charted_models.csv	text_roles	Roles of text on the model's routes
charted_models.csv	lifecycle_phases	Lifecycle phases covered by the routes (pretraining, fine-tuning, inference, evaluation)
charted_models.csv	biological_and_text_modalities	Modalities that enter the model
included_works.csv	study	First author and year of each of the work's records, as in the article's reference list
included_works.csv	proposed_models	Models proposed by the work, as charted
included_works.csv	record_ids	Identifiers of the work's records in the review database (two for works with two reports)
included_works.csv	paper_titles	Titles of the work's records
included_works.csv	dois	DOIs of the work's records
included_works.csv	year	Publication year of the work's records
included_works.csv	design_culture	Design culture of the work: token-native, encoder-mediated, hybrid, language-conditioned or alignment-only
included_works.csv	tests_dependence_on_biological_input	Whether the work reports any test of whether outputs depend on the biological input (removal, mismatch or shuffling, random input, a no-biology baseline, content perturbation, order or value ablation)
included_works.csv	uses_llm_written_text	Whether text written by an LLM was used for training or evaluation
included_works.csv	language_model_frozen	Whether the language model is kept frozen in at least one setting
included_works.csv	language_model_lora_or_adapter	Whether the language model is adapted by low-rank adaptation (LoRA) or adapters
included_works.csv	language_model_full_fine_tuning	Whether the language model is fine-tuned in full
included_works.csv	language_model_trained_from_scratch	Whether the language model is trained from scratch
included_works.csv	rationale_unchanged_architecture	Token-native work justifies its own representation by needing no change to the language model's architecture
included_works.csv	rationale_language_model_knowledge	Token-native work justifies its own representation by the language model's prior knowledge of named entities
included_works.csv	rationale_concedes_information_loss	Token-native work concedes loss of quantitative or fine-grained information in its own representation
included_works.csv	rationale_criticises_text_rendering	Encoder-mediated work criticises text rendering for losing magnitudes, low-abundance genes or structure
included_works.csv	cross_culture_comparison_units	Number of the work's comparison units between arms of different cultures
included_works.csv	opaque_step_share	Median share of the steps of the work's biological routes that are learned modules left unexpanded (empty if no biological route was decomposed)
included_works.csv	unstated_parameter_slot_share	Share of the parameter slots of the work's proposed and component models whose value is not stated in the paper (empty if there are no slots)
included_works.csv	parameter_slots	Number of parameter slots of the work's proposed and component models
motif_library.csv	motif	Name of the motif
motif_library.csv	family	Family of the motif: carrier formation, encoding or joining
motif_library.csv	operations	Library operations of the motif, in order
motif_library.csv	input_role	Role of the inputs of the operations: biological input, text input, or joining both
motif_library.csv	contiguous	Whether the operations must be adjacent (yes) or may have other operations between them (no)
motif_library.csv	papers	Number of papers with the motif
motif_library.csv	models	Number of models with the motif
motif_library.csv	pilot_papers	Number of the four author-curated pilot papers with the motif
motif_library.csv	description	What the motif does
motif_library.csv	unstated_parameter_shares	For each parameter of the motif's operations, the share of calls whose value the paper does not state
motif_library.csv	example_model	Model of the example
motif_library.csv	example_record_id	Record of the example in the review database
motif_library.csv	example_route_id	Charted route of the example
motif_library.csv	example_steps	Steps of the example: operation and the charted wording of the step
motif_library.csv	example_evidence	Verbatim quote from the paper supporting the example
gene_budgets.csv	study	First author and year of the record, as in the article's reference list
gene_budgets.csv	model_and_context	Model and the setting in which the budget applies
gene_budgets.csv	record_id	Record in the review database
gene_budgets.csv	route	Kind of input route: text or embedding, and whose input it is
gene_budgets.csv	genes_kept	Number of genes or features kept per cell or object

File	Column	Description
gene_budgets.csv	values_kept	Whether measured values are kept, and how
gene_budgets.csv	evidence	Verbatim quote from the paper
token_budgets.csv	study	First author and year of the record, as in the article's reference list
token_budgets.csv	model	Model and, where needed, the setting
token_budgets.csv	record_id	Record in the review database
token_budgets.csv	unit	What is counted, in the paper's terms
token_budgets.csv	number	Number per cell, profile or object
token_budgets.csv	evidence	Verbatim quote from the paper
llm_written_text.csv	study	First author and year of the work's records, as in the article's reference list
llm_written_text.csv	proposed_models	Models proposed by the work
llm_written_text.csv	record_ids	Records of the work in the review database
llm_written_text.csv	design_culture	Design culture of the work
llm_written_text.csv	year	Publication year
llm_written_text.csv	what_was_written	What the LLM wrote: training or evaluation content, instruction templates only, or unclear
llm_written_text.csv	generators	LLMs that wrote the text, as named in the paper
llm_written_text.csv	human_check	Whether the paper reports a human check of the generated text
llm_written_text.csv	reasoning_traces	Whether the generated text includes reasoning traces
llm_written_text.csv	llm_text_in_evaluation	Whether LLM-written text is used in the evaluation
llm_written_text.csv	example	Verbatim quote from the paper
llm_written_text.csv	how_found	How the use was identified
comparison_units.csv	unit	Identifier of the comparison unit
comparison_units.csv	study	First author and year of the work's records, as in the article's reference list
comparison_units.csv	proposed_models	Models proposed by the work that reports the comparison
comparison_units.csv	record_id	Record of the work in the review database
comparison_units.csv	task_family	Family of the task compared
comparison_units.csv	task	Task and arms, as adjudicated
comparison_units.csv	design	Design of the comparison: controlled ablation within one system, same backbone given the data as text, or different systems
comparison_units.csv	arm_a	Culture of the first arm (image rendering: DNA drawn as an image)
comparison_units.csv	arm_b	Culture of the second arm
comparison_units.csv	proposed_arm	Culture of the arm proposed by the authors
comparison_units.csv	opponent_arm	Culture of the opponent arm
comparison_units.csv	opponent_obtained_as	How the opponent arm was obtained
comparison_units.csv	winner	Culture of the winning arm, or mixed
comparison_units.csv	proposed_arm_won	Whether the proposed arm won
comparison_units.csv	token_native_arm_content	What the token-native arm writes into the token stream
comparison_units.csv	values_reported	Values reported for the comparison, as quoted in the charting (empty if none in the text)
comparison_units.csv	adjudication_note	Note on a change made at adjudication
comparison_units.csv	charting_items	Charting items merged into the unit (run and item number)
route_codebook.csv	record_id	Record in the review database
route_codebook.csv	model_id	Identifier of the charted model
route_codebook.csv	model	Model name as charted
route_codebook.csv	configuration_id	Identifier of the charted configuration
route_codebook.csv	route_id	Identifier of the charted route
route_codebook.csv	route_label	Label of the route, as charted
route_codebook.csv	lifecycle_phase	Lifecycle phase of the route
route_codebook.csv	source_modality	Modality of the route's source object
route_codebook.csv	carrier_family	Family of the carrier that reaches the model
route_codebook.csv	carrier_subtype	Subtype of the carrier
route_codebook.csv	fusion_topology	How the carrier is fused with the language model
route_codebook.csv	text_role	Role of text on the route
route_codebook.csv	operation_chain	Consensus library operations of the route, in order
route_codebook.csv	carrier_formation	Design axis 1, carrier formation: verbalised, symbolic, embedded, perceptual or text source
route_codebook.csv	carrier_formation_rule	Rule that assigned the carrier formation
route_codebook.csv	interface	Design axis 2, interface to the language model
route_codebook.csv	interface_rule	Rule that assigned the interface
route_codebook.csv	value_transformation	Design axis 3, transformation of measured values: verbatim, order only, discretised, learned compression, or not applicable

File	Column	Description
route_codebook.csv	value_transformation_rule	Rule that assigned the value transformation
route_codebook.csv	value_transformation_ modifiers	Modifiers of the value transformation: subset, normalised
route_codebook.csv	value_transformation_ evidence_steps	Identifiers of the decomposed steps on which the value transformation rests
route_codebook.csv	text_role_axis	Design axis 4, text role: instruction, payload, supervision, context or none
route_codebook.csv	transparency	Design axis 5, transparency: transparent, partial, opaque or no computation
route_codebook.csv	opaque_step_share	Share of the route's steps that are learned modules left unexpanded
route_codebook.csv	stated_parameter_share	Share of the route's parameter slots whose value the paper states
route_codebook.csv	archetype	Carrier × interface archetype of the route